

# DSC-W300

Ver. 1.3 2008.08

Revision History

## SECTION 6 ADJUSTMENTS

Auto-ADJ

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- Use this Service Manual together with the Automatic Adjustment Program (DSC-W300 Auto-Adj Ver\_1.3r04.exe).

# SONY®

## SECTION 6 ADJUSTMENTS

### Before starting adjustments

#### 1-1. Adjusting items when replacing main parts and boards

When replacing main parts and boards, adjust the items indicated by ● in the following table.

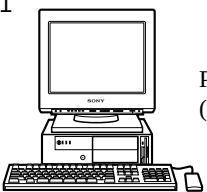
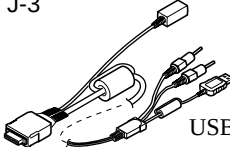
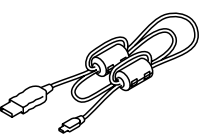
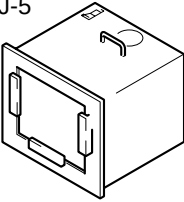
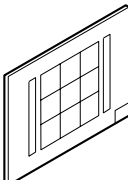
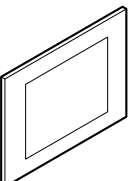
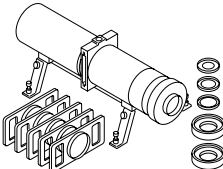
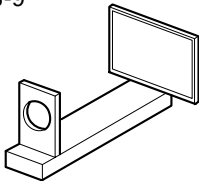
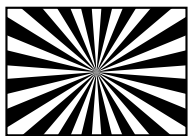
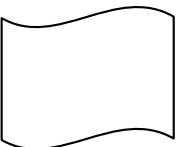
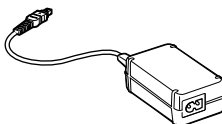
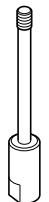
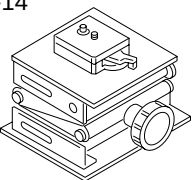
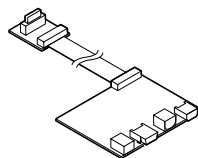
**Note:** When replacing the SY-199 board, erase the data in internal memory of the board before replacement.

| Adjusting item                   | Adjustment                               | Replaced parts    |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
|----------------------------------|------------------------------------------|-------------------|-----------------------------------------------------------------|------------|---------------------------|--------|----------|-------------------|------|--------------|-------|--------------|-------|--------------|--------------|--------------|--------------|--------------|
|                                  |                                          | Block replacement |                                                                 |            | Mounted parts replacement |        |          | Board replacement |      |              |       |              |       |              |              |              |              |              |
|                                  |                                          | Lens block        | CCD block assy (Including CD-734 flexible board and CCD imager) | Flash unit | LCD block                 | LCD901 | LCD unit | AF-116 board      | D001 | SY-199 board | IC301 | SY-199 board | IC602 | SY-199 board | SE502, SE501 | AF-116 board | ST-191 board | SY-199 board |
|                                  |                                          |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
| Destination Data Write           | Destination data write                   |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              | ●            |
| USB Serial No. Input             | USB serial No. input                     |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              | ●            |
| Composite video level adjustment | Composite video level adj.               |                   |                                                                 |            |                           |        |          |                   |      |              |       | ●            |       |              |              |              |              | ●            |
| Component video level adjustment | Component out Y level adj.               |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
|                                  | Component out Pb level adj.              |                   |                                                                 |            |                           |        |          |                   |      |              |       | ●            |       |              |              |              |              | ●            |
|                                  | Component out Pr level adj.              |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
| CAMERA adjustment 1              | Hall adj.                                | ●                 |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              | ●            |
|                                  | Auto orientation adj.                    |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
| CAMERA adjustment 2              | Flange back adj.                         | ●                 | ●                                                               |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              | ●            |
| CAMERA adjustment 3              | Flange back check                        | ●                 | ●                                                               |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              | ●            |
| CAMERA adjustment 4              | F No. compensation                       |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
|                                  | Measure gain, LV adj.                    | ●                 | ●                                                               |            |                           |        |          |                   |      | ●            |       |              |       |              |              |              |              | ●            |
|                                  | Mechanical shutter adj.                  |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
| CAMERA adjustment 5              | AWB 3200K-5800K standard data input      |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
|                                  | AWB 3200K-5800K check                    |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
|                                  | Color reproduction adj. & check          |                   | ●                                                               |            |                           |        |          |                   |      | ●            |       |              |       |              |              |              |              | ●            |
|                                  | CCD white defect compensation check      |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
|                                  | CCD black defect compensation check      |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |
| CAMERA adjustment 6              | Strobe adj.                              |                   | ●                                                               | ●          |                           |        |          |                   | ●    |              |       |              |       |              |              |              | ●            | ●            |
| CAMERA adjustment 7              | Auto focus illumination check            |                   |                                                                 |            |                           |        |          | ●                 |      |              |       |              |       |              | ●            |              |              | ●            |
| CAMERA adjustment 8              | Angular velocity sensor sensitivity adj. | ●                 |                                                                 |            |                           |        |          |                   |      |              |       |              | ●     |              |              |              |              | ●            |
| LCD adjustment                   | V-COM adj.                               |                   |                                                                 |            |                           | ●      |          |                   |      |              |       |              |       |              |              |              |              | ●            |
|                                  | White Balance adj.                       |                   |                                                                 |            |                           |        |          |                   |      |              |       |              |       |              |              |              |              |              |

**Table 6-1-1**

## 1-2. List of service tools

- Oscilloscope
- Color monitor

|                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                        |                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>J-1</p>  <p>Personal computer<br/>(Note 1)</p>                                                         | <p>J-2</p>  <p>HASP key and application<br/>for adjustment (SeusEX)</p> <p>Contact our service headquarters of each<br/>area how to get the application for<br/>adjustment (SeusEX) and HASP key.</p> | <p>J-3</p>  <p>USB, A/V, DC IN cable<br/>for multi-use terminal<br/>1-830-848-21</p> |
| <p>J-4</p>  <p>USB cable<br/>1-829-868-41</p>                                                             | <p>J-5</p>  <p>Pattern box PTB-450<br/>J-6082-200-A<br/>or<br/>Small pattern box<br/>PTB-1450<br/>J-6082-557-A</p>                                                                                    | <p>J-6</p>  <p>9 colors chart (Note 2)</p> <p>For PTB-1450:<br/>J-6082-562-A</p>     |
| <p>J-7</p>  <p>Clear chart</p> <p>For PTB-450:<br/>J-6080-621-A</p> <p>For PTB-1450:<br/>J-6082-560-A</p> | <p>J-8</p>  <p>Minipattern box<br/>J-6082-353-B</p>                                                                                                                                                   | <p>J-9</p>  <p>Flange back<br/>adjustment jig<br/>J-6082-563-A</p>                   |
| <p>J-10</p>  <p>Siemens star chart<br/>J-6080-875-A</p>                                                  | <p>J-11</p>  <p>Background paper<br/>J-2501-130-A</p>                                                                                                                                                | <p>J-12</p>  <p>AC power adaptor<br/>AC-LS5<br/>1-479-284-51</p>                    |
| <p>J-13</p>  <p>Tripod Adaptor<br/>J-6082-577-B</p>                                                     | <p>J-14</p>  <p>Camera table<br/>J-6082-384-A</p>                                                                                                                                                   | <p>J-15</p>  <p>Component<br/>video out jig<br/>J-6082-659-A</p>                   |

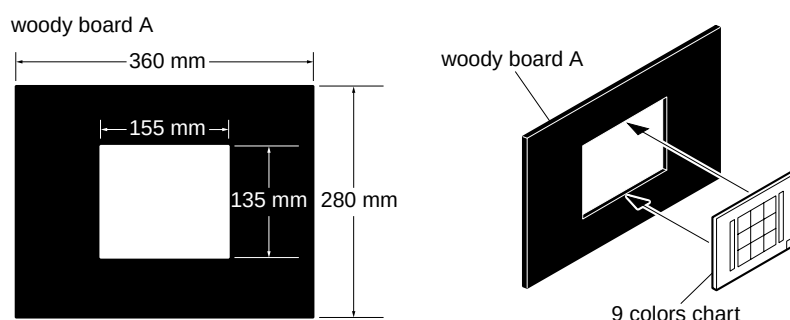
**Fig. 6-1-1**

### Note 1: Personal computer

OS: Windows 2000/XP Home/XP Pro  
RAM: 256 MB or more recommended  
USB: 2.0 recommended (also compatible with 1.1)  
Two connectors are required.

### Note 2: In using the 9 colors chart on the pattern box PTB-450, adjust the chart size through the procedure shown below so that it matches to the pattern box PTB-450.

- 1) Prepare a woody board A of the thickness 5 mm, and paint it mat-black.
- 2) Fit the 9 colors chart in the woody board A, and secure the chart with a black tape, etc. to shield the light.



## 6-1. CAMERA SECTION ADJUSTMENTS

### 1-1. PREPARATIONS BEFORE ADJUSTMENTS

#### 1-1-1. Preparations

- 1) Connect the equipment for adjustments according to Fig. 6-1-2.
- 2) Start up the application for adjustment (SeusEX).

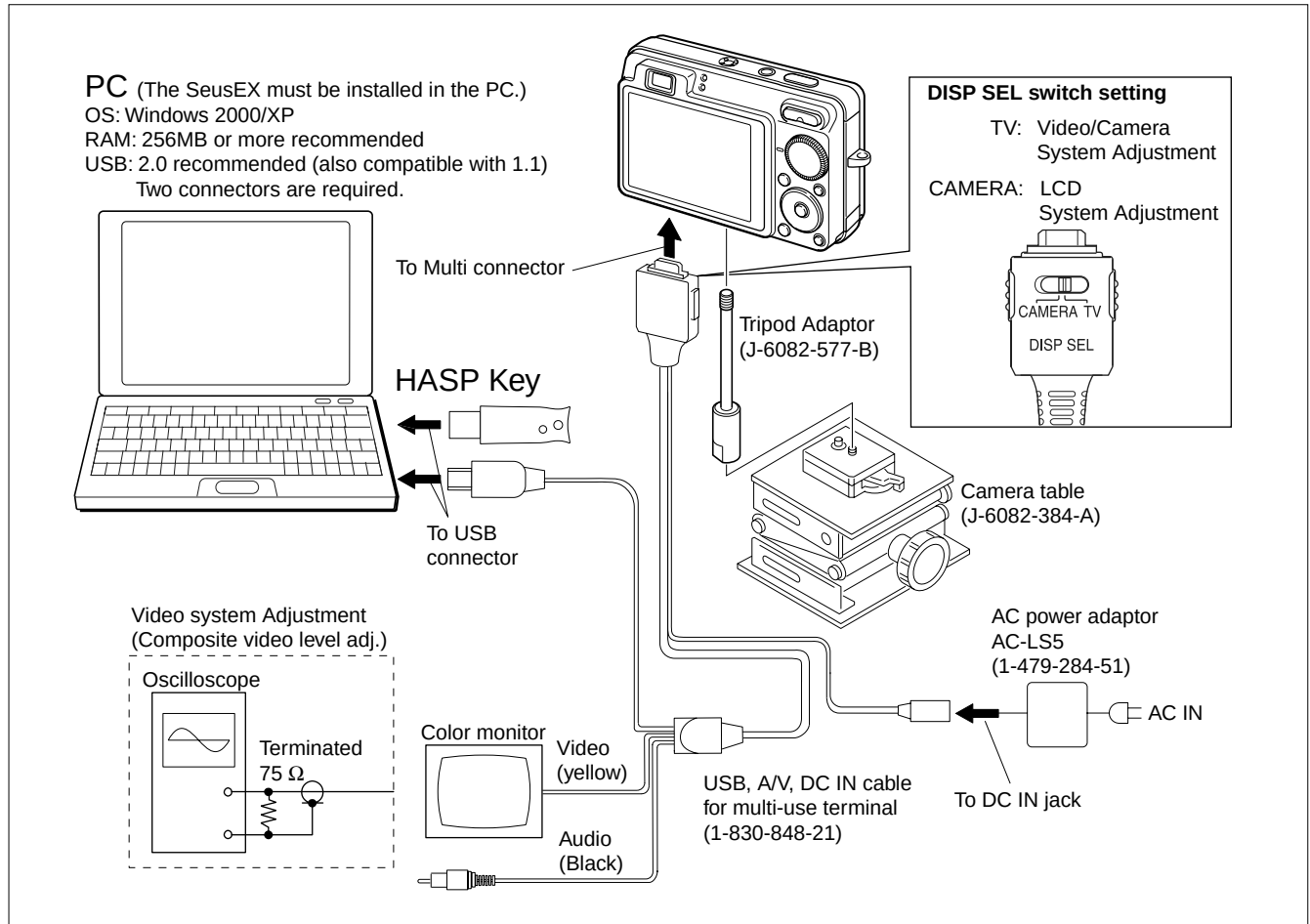
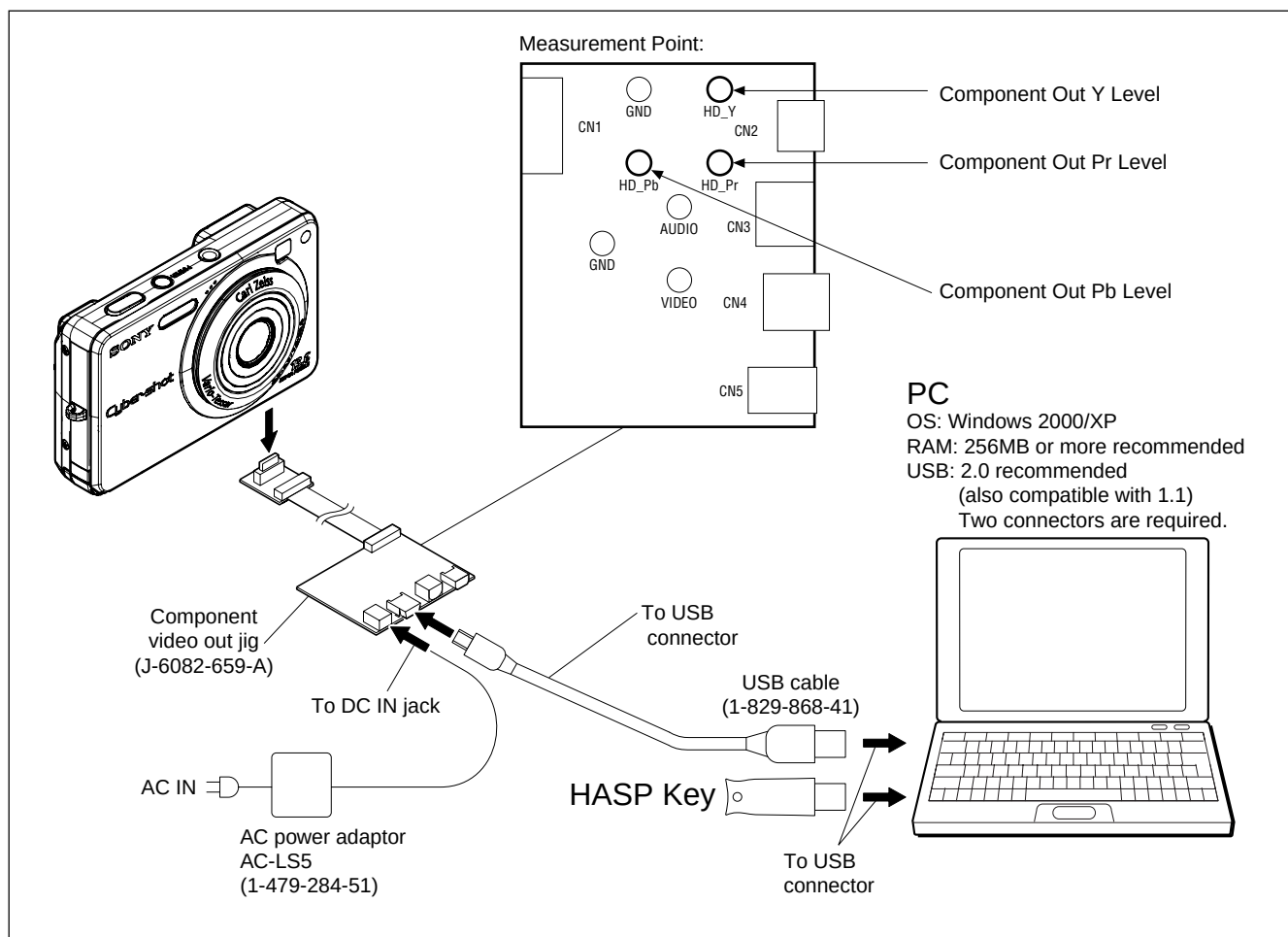


Fig. 6-1-2

**Note:** The set must be connected to the Component video out jig when performing the “Component out (Y, Pb, Pr) level adjustment” in the “Video System Adjustment”.

- Connection diagram when “Component out (Y, Pb, Pr) level adjustment” in “Video System Adjustment” is performed.



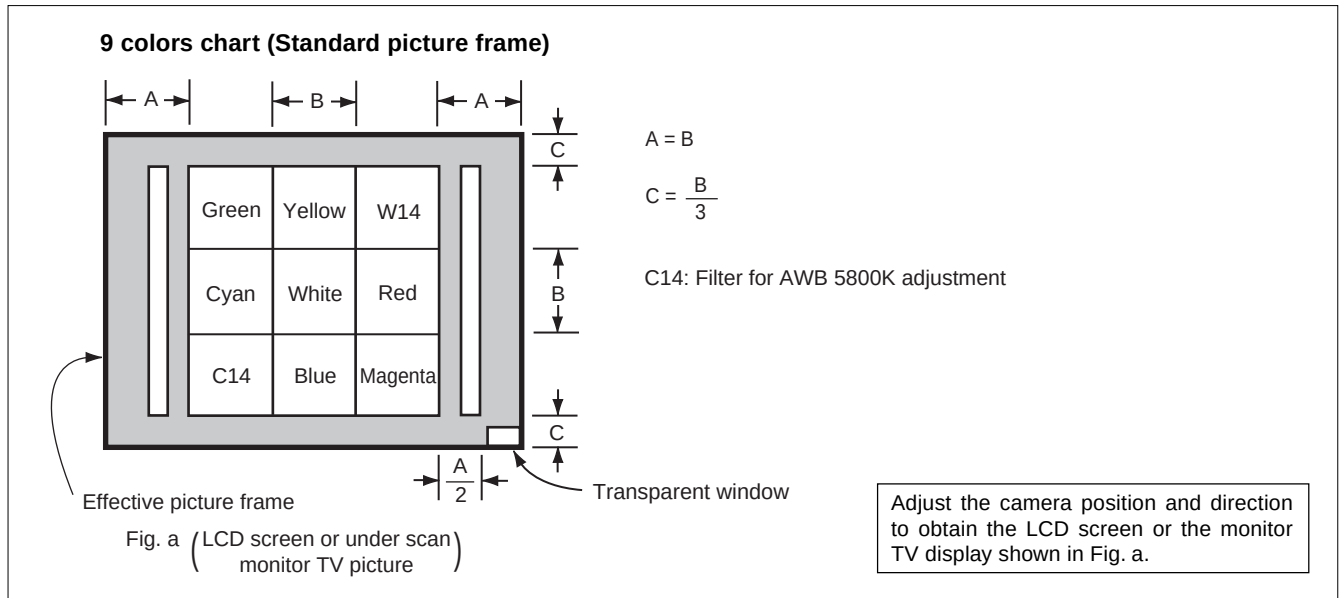
**Fig. 6-1-3**

## 1-1-2. Precautions

### 1. Setting the Switch

Unless otherwise specified, set the switches as follows and perform adjustments.

1. Mode dial ..... Auto Adjustment
2. W/T (Zoom) button ..... WIDE end
3. Digital Zoom  
(Settings – Shooting Settings – Shooting Settings 1) .... Off



**Fig. 6-1-4**

### 2. Subjects

- 1) 9 colors chart (Standard picture frame).

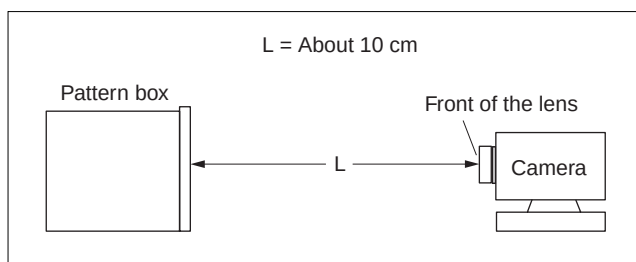
When performing adjustments using the 9 colors chart, adjust the picture frame as shown in Fig. 6-1-4. (Standard picture frame)

- 2) Clear chart (Standard picture frame)

Remove the 9 colors chart from the pattern box and insert a clear chart in its place. (Do not perform zoom operations during this time)

### 3. Setting Distance between Pattern Box and Camera

Set the distance from the front of the lens to the pattern box as shown in Fig. 6-1-5.



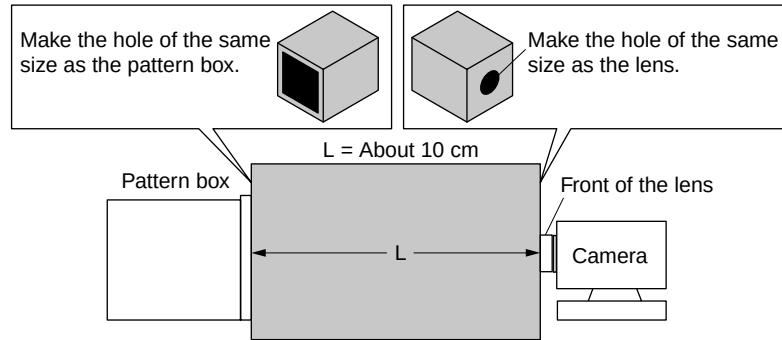
**Fig. 6-1-5**

#### 4. Precautions When Using Pattern Box

- 1) It takes about 30 minutes for pattern box to stabilize its brightness.  
Turn on the pattern box 30 minutes before the adjustment starts.
- 2) Make arrangement so that the outside light does not enter the chart surface in the pattern box.  
Also, place a board between chart and camera, and make a hole at the lens part of the board so that the camera is not reflected in the shot image screen.

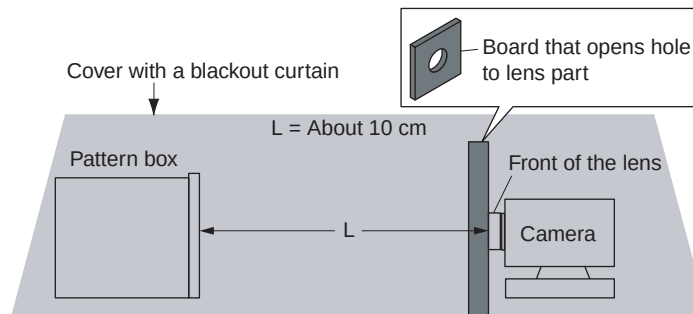
(Adjustment may not be performed correctly due to the influence of outside light.)

**Example 1:** Place a box to block a section between pattern box and camera.



**Fig. 6-1-6**

**Example 2:** Place a board having a hole in front of the camera and cover the pattern box and camera with a blackout curtain



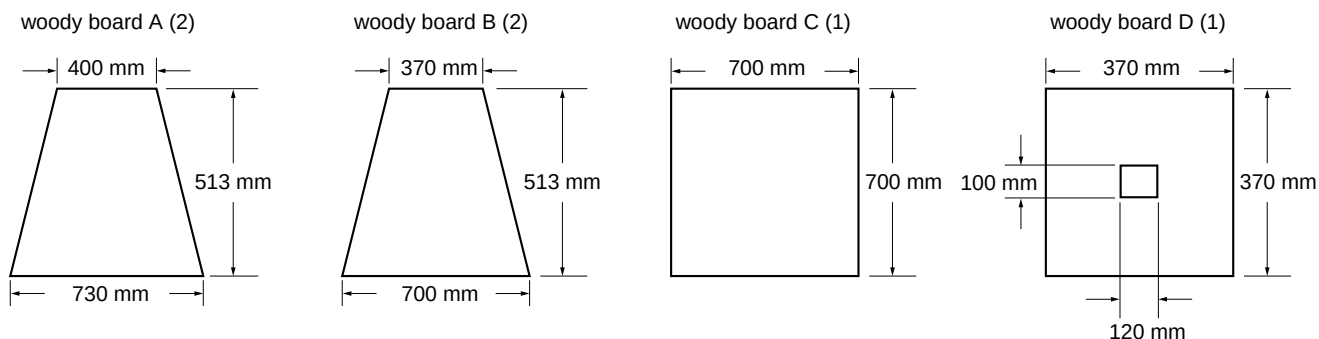
**Fig. 6-1-7**

- 3) Control of color chart  
The color chart will fade if it is exposed to direct sunlight or strong light.  
Since the fading of color chart progresses even with the light in the pattern box, remove and store the color chart when it is not used.  
Remove the color chart and store it.  
Store the color chart in a place not exposed to direct light, avoiding high temperature and humidity.  
Use the color chart for about three years, and afterward replace it with a new chart.

## 5. Preparing the Flash Adjustment Box

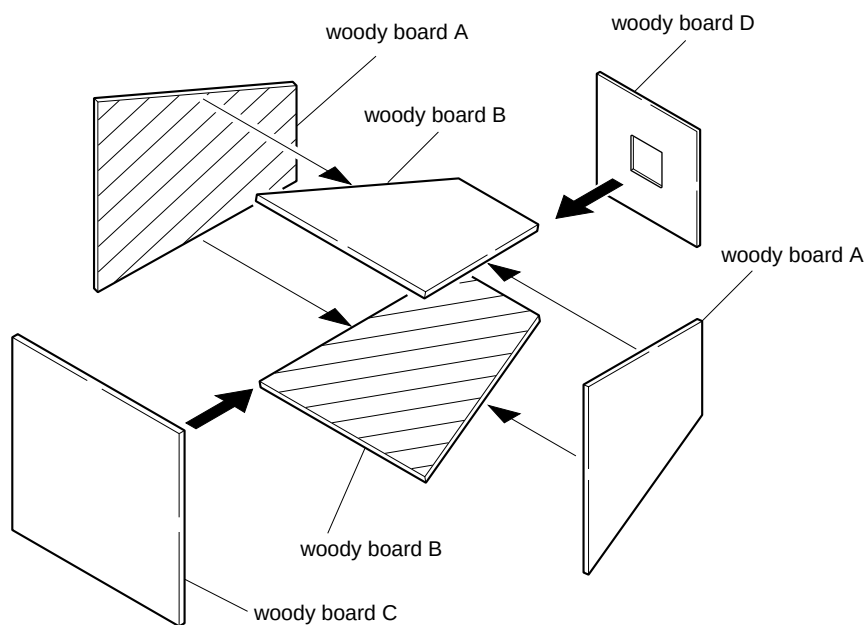
A dark room is required to provide an accurate flash adjustment. If it is not available, prepare the flash adjustment box as given below;

- 1) Provide woody board A, B, C and D of 15 mm thickness.

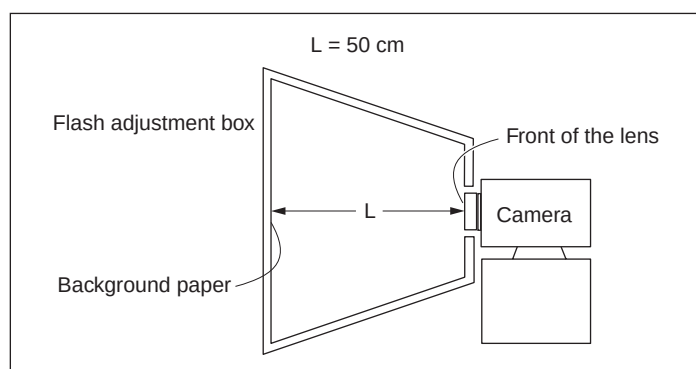


**Fig. 6-1-8**

- 2) Apply black mat paint to one side of woody board A, B and D.
- 3) Attach background paper (J-2501-130-A) to woody board C.
- 4) Assemble so that the black sides and the background paper side of woody board A, B, C and D are internal. (Fig. 6-1-9)



**Fig. 6-1-9**



**Fig. 6-1-10**

## 1-2. ADJUSTMENT PROGRAM

The DSC-W300 is adjusted by the Automatic Adjustment Program. The Automatic Adjustment Program enters automatically via the SeusEX the adjustment operations that were formerly entered manually by the adjustment remote commander (some items may be adjusted by manual operation on the operation screen of the SeusEX).

### 1. Precautions When Using Automatic Adjustment Program

- 1) The Automatic Adjustment Program writes the adjustment results such as EVR data to the set through two-way communication with the camera via the SeusEX. Accordingly, the Automatic Adjustment Program must be used in the environment where the SeusEX operates.
- 2) The Automatic Adjustment Program cannot be used when the SEUS or the SeusCam is running. Exit the SEUS or the SeusCam before using the Automatic Adjustment Program.
- 3) The SeusEX must be already started on the PC when using the Automatic Adjustment Program. With the SeusEX not started, some adjustment items will take time in adjustment.
- 4) The program run time may vary depending on the environment of the personal computer used.

### 2. Start of Automatic Adjustment Program

Double-click the application file (DSC-W300 Auto-Adj Ver\_1.3r04.exe), and the Automatic Adjustment Program will start.

### 3. Function of Each Button on Main Menu Screen

When the Automatic Adjustment Program started, the Main Menu screen in Fig. 6-1-11 will appear. On this screen, select each adjustment section.

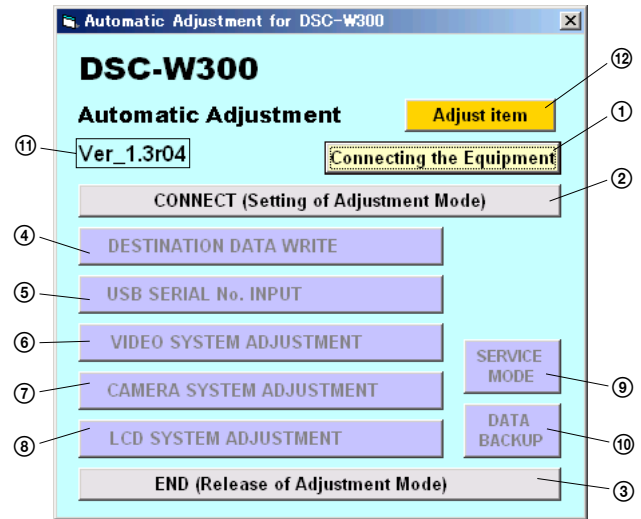


Fig. 6-1-11

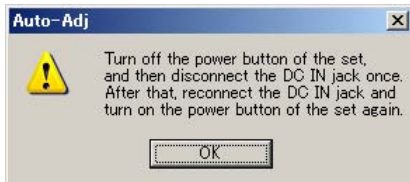
- ① **Connecting the Equipment** button  
A connection diagram of the equipment is displayed.
- ② **CONNECT** button  
The mode of Camcorder is switched to the Adjustment Mode. When the Adjustment Mode has switched normally, the operation of the buttons ④ – ⑩ is enabled.
- ③ **END** button  
The mode of Camcorder is switched to the normal mode. When the normal mode has switched correctly, the Automatic Adjustment Program is finished.
- ④ **DESTINATION DATA WRITE** button  
The “DESTINATION DATA WRITE” screen appears.
- ⑤ **USB SERIAL NO. INPUT** button  
The “USB SERIAL No. INPUT” screen appears.
- ⑥ **VIDEO SYSTEM ADJUSTMENT** button  
The “VIDEO SYSTEM ADJUSTMENT” screen appears.
- ⑦ **CAMERA SYSTEM ADJUSTMENT** button  
The “CAMERA SYSTEM ADJUSTMENT” screen appears.
- ⑧ **LCD SYSTEM ADJUSTMENT** button  
The “LCD SYSTEM ADJUSTMENT” screen appears.
- ⑨ **SERVICE MODE** button  
The “SERVICE MODE” screen appears.
- ⑩ **DATA BACKUP** button  
The “DATA BACKUP” screen appears.
- ⑪ This part indicates the version of Automatic Adjustment Program.
- ⑫ **Adjust item** button  
“Adjusting items when replacing main parts and boards” table is displayed.

#### 4. Setting of Adjustment Mode

Before performing the adjustment, "Setting of Adjustment Mode" is required.

##### [Setting method]

- 1) Connect the Camera to the PC with a USB cable, and turn on the power switch.
- 2) Start the Automatic Adjustment Program, and click the **[Connect]** button on the Main Menu screen.
- 3) Turn off the power button of the set when the following message is displayed.



- 4) Once turn off a set completely by pulling a "DC In Jack" (Or you can pull a Multi-connector).
- 5) Connect a "DC In Jack" again and turn on a set by the power button of a body.  
Click the **[OK]** button on the message screen when the set started.
- 6) Upon successful completion of the settings in the Adjustment Mode, the operation of each button on the Main Menu screen is enabled.

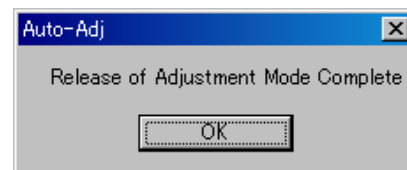
#### 5. Release of Adjustment Mode

To finish the adjustment, be sure to perform "Release of Adjustment Mode".

##### [Releasing method]

- 1) Click the **[END]** button on the Main Menu screen.
- 2) When the following message is displayed, releasing of adjustment mode has completed. Click the **[OK]** button in the message window to exit the Automatic Adjustment Program.

**Note:** The Camera switches to the normal mode by turning off and on the power switch. After the adjustment finished, turn off and on again the power switch of the Camera to confirm that the USB mode screen is displayed.



### 1-3. DESTINATION DATA WRITE

**Note:** The DESTINATION DATA WRITE cannot be set with other than the Service board.

#### 1. Function of Each Button on Destination Data Write Screen

Click the [DESTINATION DATA WRITE] button on the Main Menu screen, and the “DESTINATION DATA WRITE” screen in Fig. 6-1-12 will appear.

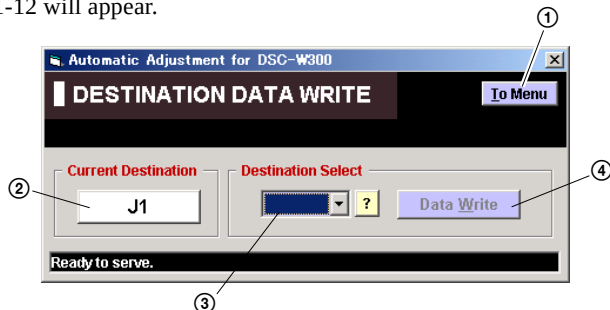


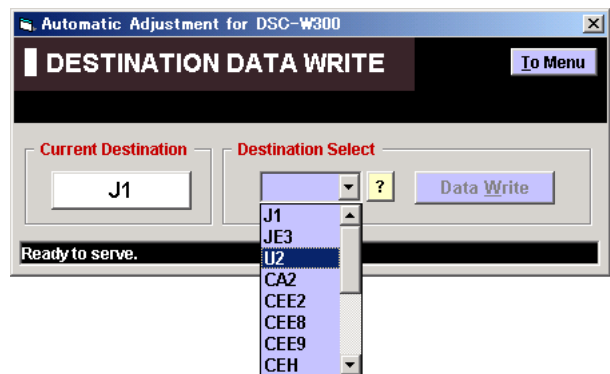
Fig. 6-1-12

- ① [To Menu] button  
Return to the main menu.
- ② Destination Check button  
Current destination setting checked when the “DESTINATION DATA WRITE” screen started is displayed.  
When this button is clicked, the destination is checked and the display is updated.
- ③ Destination List  
Select the written destination.
- ④ [Data Write] button  
Write the destination data to the camera.

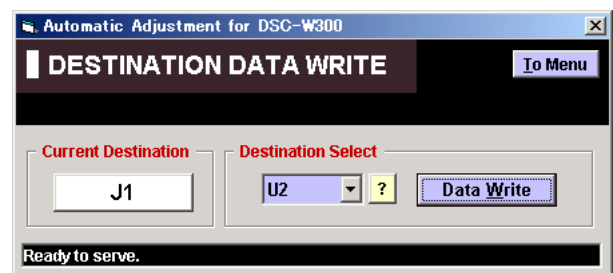
#### 2. Destination Data Write

[Writing method]

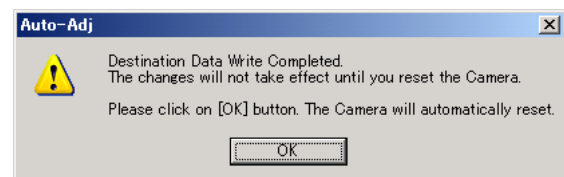
- 1) Select the written destination with the Destination list.



- 2) Click the [Data Write] button.



- 3) Following message will be appeared after completing data writing.



- 4) After the destination data writing completed, click the Destination Check button to check the destination

### 3. Selectable Language Table

| DESTINATION | AREA | SELECTABLE LANGUAGE |         |        |        |         |         |            |                    |                     |       |         |        |         |        |      |        |         |           |        |         |        |       |           |         | VIDEO OUT Default |       |
|-------------|------|---------------------|---------|--------|--------|---------|---------|------------|--------------------|---------------------|-------|---------|--------|---------|--------|------|--------|---------|-----------|--------|---------|--------|-------|-----------|---------|-------------------|-------|
|             |      | Japanese            | English | French | German | Spanish | Italian | Portuguese | Simplified Chinese | Traditional Chinese | Dutch | Russian | Korean | Persian | Arabic | Thai | Melayu | Swedish | Norwegian | Danish | Finnish | Polish | Czech | Hungarian | Turkish |                   | Greek |
| J1          | J    | ●                   |         |        |        |         |         |            |                    |                     |       |         |        |         |        |      |        |         |           |        |         |        |       |           |         | NTSC              |       |
| JE3         | JE   |                     | ●       |        |        | ○       |         | ○          | ○                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         | PAL               |       |
| U2          | US   |                     | ●       | ○      |        | ○       | ○       |            | ○                  | ○                   |       |         |        |         |        |      |        |         |           |        |         |        |       |           |         | NTSC              |       |
| CA2         | CND  |                     | ●       | ○      |        | ○       | ○       |            | ○                  | ○                   |       |         |        |         |        |      |        |         |           |        |         |        |       |           |         | NTSC              |       |
| CEE2        | AEP  |                     | ○       | ○      | ○      | ○       | ○       | ○          |                    |                     | ○     | ●       |        |         |        |      |        | ○       | ○         | ○      | ○       | ○      | ○     | ○         | ○       | ○                 | PAL   |
| CEE8        |      |                     | ●       | ○      | ○      | ○       | ○       | ○          |                    |                     | ○     |         |        |         |        |      |        | ○       | ○         | ○      | ○       | ○      | ○     | ○         | ○       | ○                 | PAL   |
| CEE9        |      |                     | ●       | ○      | ○      | ○       | ○       | ○          |                    |                     | ○     | ○       |        |         |        |      |        | ○       | ○         | ○      | ○       | ○      | ○     | ○         | ○       | ○                 | PAL   |
| CEH         | UK   |                     | ●       | ○      | ○      | ○       | ○       | ○          |                    |                     | ○     |         |        |         |        |      | ○      | ○       | ○         | ○      | ○       | ○      | ○     | ○         | ○       | ○                 | PAL   |
| E15         | E    |                     | ●       |        |        | ○       |         | ○          | ○                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | PAL   |
| E32         |      |                     | ●       |        |        | ○       |         | ○          | ○                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | PAL   |
| E33         |      |                     | ○       |        |        | ●       |         | ○          | ○                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | NTSC  |
| TH6         | Thai |                     | ○       |        |        | ○       |         | ○          | ○                  | ○                   |       |         | ○      | ○       |        | ●    | ○      |         |           |        |         |        |       |           |         |                   | PAL   |
| AU2         | AUS  |                     | ●       | ○      |        | ○       | ○       |            | ○                  | ○                   |       |         |        |         |        |      |        |         |           |        |         |        |       |           |         |                   | PAL   |
| HK1         | HK   |                     | ●       |        |        | ○       |         | ○          | ○                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | PAL   |
| CN2         | CH   |                     | ○       |        |        | ○       |         | ○          | ●                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | PAL   |
| KR2         | KR   |                     | ○       |        |        | ○       |         | ○          | ○                  | ○                   |       |         | ●      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | NTSC  |
| AR2         | AR   |                     | ○       |        |        | ●       |         | ○          | ○                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | NTSC  |
| BR1         | BR   |                     | ○       |        |        | ○       |         | ●          | ○                  | ○                   |       |         | ○      | ○       | ○      | ○    | ○      |         |           |        |         |        |       |           |         |                   | NTSC  |

●: INITIAL LANGUAGE

Table 6-1-2

## 1-4. USB SERIAL No. INPUT

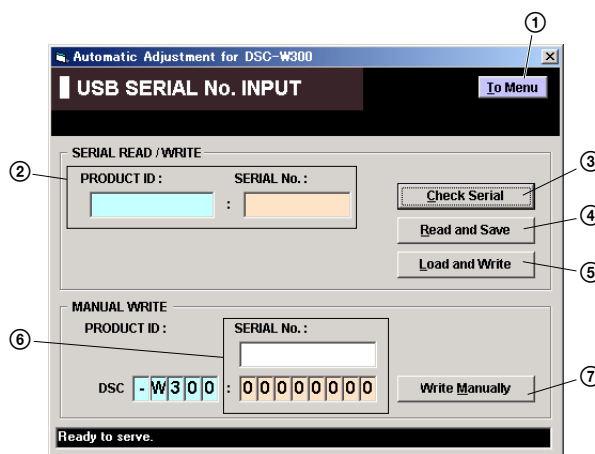
The set is shipped with a unique ID (USB Serial No.) written in it. This ID has not been written in a new board for service, and therefore it must be entered after the board replacement.

If original ID can be read from the board before replacement, read it from the board before replacement using the “SERIAL READ/ WRITE” screen, and then write it after replacement.

If original ID cannot be read from the board before replacement, write the ID for service using the “MANUAL WRITE” screen. (The ID for service is different from the ID written when the set is shipped.) Enter the PRODUCT ID (last 5 characters of model name) and SERIAL No. into the screen and write them.

## 1. Function of Each Button on USB Serial No. Input Screen

Click the **USB SERIAL No. INPUT** button on the Main Menu screen, and “USB SERIAL No. INPUT” screen in Fig. 6-1-13 will appear.



**Fig. 6-1-13**

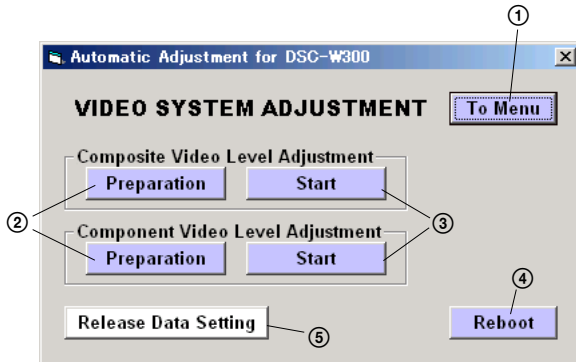
- ① **To Menu** button  
Return to the main menu.
- ② Display area  
The “PRODUCT ID” and “SERIAL No.” are displayed.
- ③ **Check Serial** button  
The USB SERIAL No. data is read from the camera and displayed in the display area.
- ④ **Read and Save** button  
The USB SERIAL No. data is read from the camera and saved in PC as a file.  
Default file name is as follows:  

DSC-W300\_SERIAL\_XXXXXXXX\_YYMMDD.dat
- ⑤ **Load and Write** button  
The USB SERIAL No. data is loaded from the file saved in PC and written to the camera.
- ⑥ Input area  
Enter the “SERIAL No.”.  
For the “PRODUCT ID”, the last 5 characters of model name are fixed. For the “SERIAL No.”, read it from the label on the camera body and enter it.
- ⑦ **Write Manually** button  
The USB SERIAL No. data entered in the input area is written to the camera.

## 1-5. VIDEO SYSTEM ADJUSTMENTS

### 1-5-1. Function of Each Button on Video System Adjustment Screen

Click the [VIDEO SYSTEM ADJUSTMENT] button on the Main Menu screen, and the “VIDEO SYSTEM ADJUSTMENT” screen in Fig. 6-1-14 will appear.



**Fig. 6-1-14**

- ① [To Menu] button  
Return to the main menu.
- ② [Preparation] button  
Notes for adjustment or jigs used are displayed.
- ③ [Start] button  
Each adjustment “Composite Video Level Adjustment” or “Component Video Level Adjustment” starts.
- ④ [Reboot] button  
When this button is clicked, the camera is rebooted.
- ⑤ [Release Data Setting] button  
The data setting at the adjustment is cancelled.  
During the data setting, the button color changes from “white” to “red”. When the data setting is cancelled, the button color returns to “white”.  
(Use this button when an error occurred in the video adjustment. If the adjustment completed successfully, the data setting is automatically cancelled and the button color returns to “white”.)

### 1-5-2. Adjustment Items of VIDEO System Adjustment

The adjustment items of video system adjustment are as listed in Table 6-1-3. The Automatic Adjustment Program executes the adjustment items if the VIDEO Adjustment Start button is clicked.

| Button Name                      | Adjustment                  | Measurement Point                                                                  | Measuring Instrument | Adjusting Address |      |         |
|----------------------------------|-----------------------------|------------------------------------------------------------------------------------|----------------------|-------------------|------|---------|
|                                  |                             |                                                                                    |                      | Block             | Page | Address |
| Composite Video Level Adjustment | Composite Video Level Adj.  | VIDEO terminal of USB, A/V, DC IN cable for multi-use terminal (75 ohm terminated) | Oscilloscope         | 11                | 60   | 06B8    |
| Component Video Level Adjustment | Component Out Y Level Adj.  | Land of HD_Y (Component video out jig)                                             | Oscilloscope         | 11                | 60   | 0680    |
|                                  | Component Out Pb Level Adj. | Land of HD_Pb (Component video out jig)                                            | Oscilloscope         | 11                | 60   | 0681    |
|                                  | Component Out Pr Level Adj. | Land of HD_Pr (Component video out jig)                                            | Oscilloscope         | 11                | 60   | 0682    |

**Table 6-1-3**

### 1-5-3. Adjusting Method

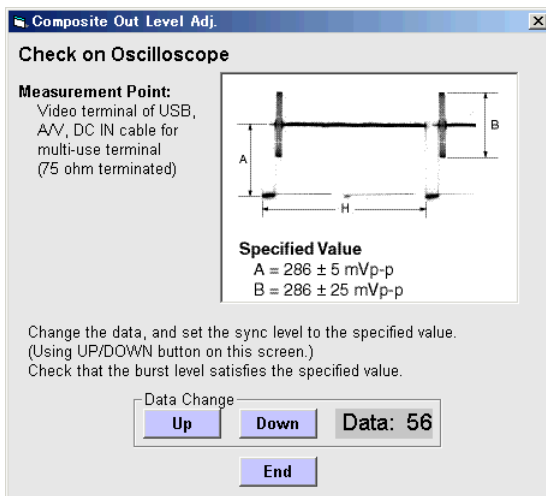
#### 1. Composite Video Level Adjustment

##### [Automatic Adjustment Program execution items and sequence]

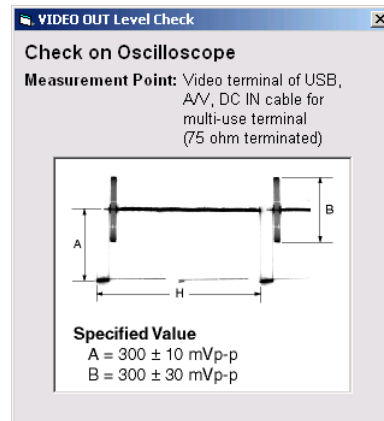
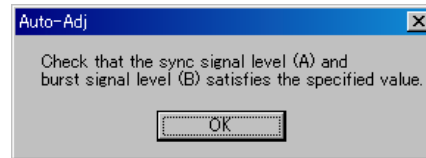
1. Data Setting during Video Adj.
2. Composite Video Level Adj.
3. Release of Data Setting during Video Adj.

##### [Adjusting method]

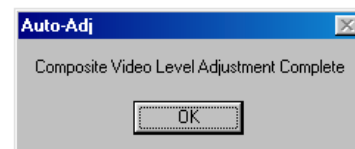
- 1) Click the **[Start]** button of the Composite Video Level Adj.
- 2) The Automatic Adjustment Program executes the “1. Data Setting during Video Adj.”.
- 3) If “1. Data Setting during Video Adj.” completed successfully, the following screen is displayed during the execution of “2. Composite Video Level Adj.”. Using the **[Up]**/**[Down]** button on the screen, adjust so that the sync level of the video signals satisfies the specified value. After the adjustment, click the **[End]** button in the screen.



- 4) If the **[End]** button is clicked, the following message and screen are displayed. Check that the sync signal level and burst level of the video signals satisfies the specified value, and click the **[OK]** button in the message.



- 5) If the **[OK]** button is clicked, “3. Release of Data Setting during Video Adj.” will be executed.
- 6) Upon successful completion of all item the Composite Video Level Adjustment, the following message is displayed. Click the **[OK]** button.



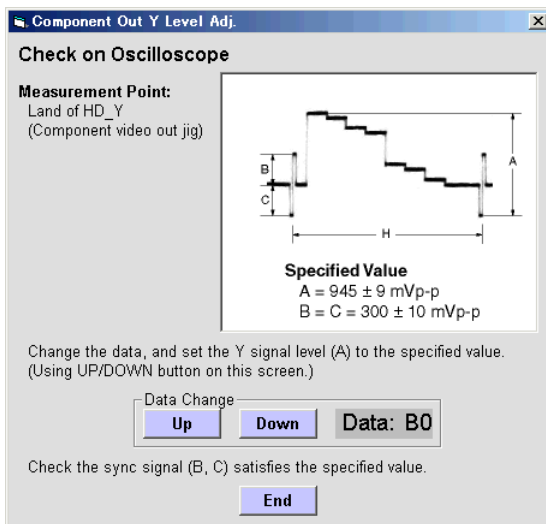
## 2. Component Video Level Adjustment

### [Automatic Adjustment Program execution items and sequence]

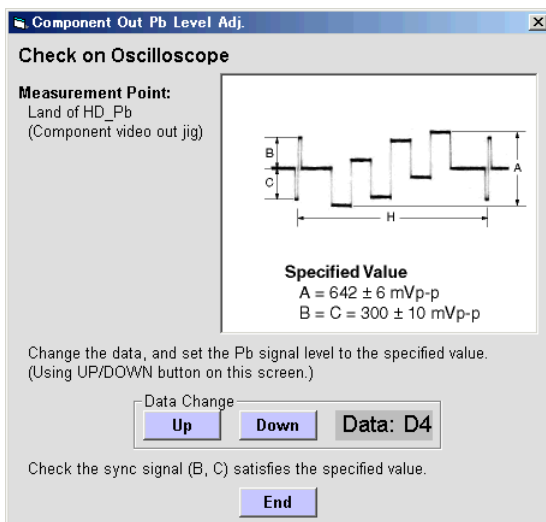
1. Data Setting during Video Adj.
2. Component Out Y Level Adj.
3. Component Out Pb Level Adj.
4. Component Out Pr Level Adj.
5. Release of Data Setting during Video Adj.

### [Adjusting method]

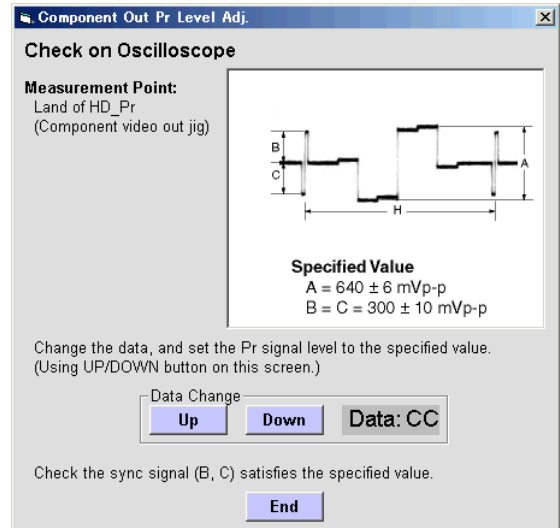
- 1) Click the **[Start]** button of the Component Video Level Adjustment.
- 2) The Automatic Adjustment Program executes the “1. Data Setting during Video Adj.”.
- 3) If “1. Data Setting during Video Adj.” completed successfully, the following screen is displayed during the execution of “2. Component Out Y Level Adj.”. Using the **[Up]/[Down]** button on the screen, adjust so that the Y signal level satisfies the specified value. After the adjustment, check that the sync level of the Y signals satisfies the specified value, and click the **[End]** button in the screen.



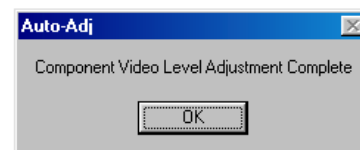
- 4) After that, the following screen is displayed during the execution of “3. Component Out Pb Level Adj.”. Using the **[Up]/[Down]** button on the screen, adjust so that the Pb signal level satisfies the specified value. After the adjustment, check that the sync level of the Pb signals satisfies the specified value, and click the **[End]** button in the screen.



- 5) After that, the following screen is displayed during the execution of “4. Component Out Pr Level Adj.”. Using the **[Up]/[Down]** button on the screen, adjust so that the Pr signal level satisfies the specified value. After the adjustment, check that the sync level of the Pr signals satisfies the specified value, and click the **[End]** button in the screen.



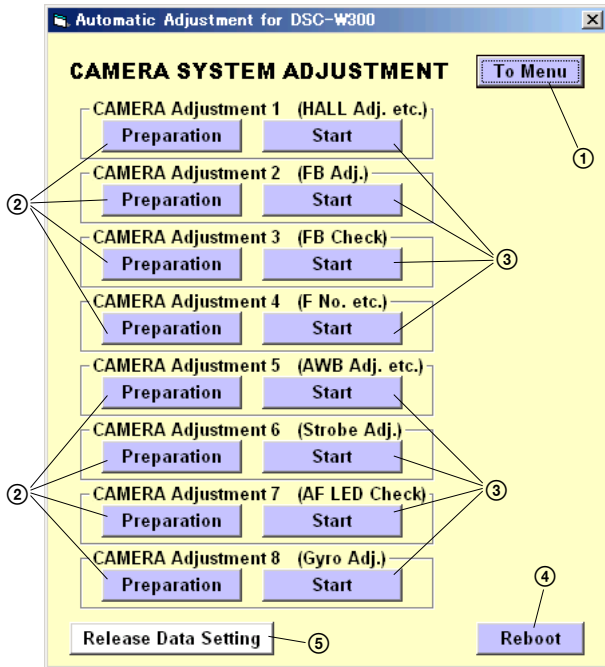
- 6) If the **[End]** button is clicked, “5. Release of Data Setting during Video Adj.” will be executed.
- 7) Upon successful completion of all item the Component Video Level Adjustment, the following message is displayed. Click the **[OK]** button.



## 1-6. CAMERA SYSTEM ADJUSTMENTS

### 1-6-1. Function of Each Button on Camera System Adjustment Screen

Click the **CAMERA SYSTEM ADJUSTMENT** button on the Main Menu screen, and the “CAMERA SYSTEM ADJUSTMENT” screen in Fig. 6-1-15 will appear.



**Fig. 6-1-15**

- ① **To Menu** button  
Return to the main menu.
- ② **Preparation** button  
Notes for adjustment or jigs used are displayed.
- ③ **Start** button  
Each adjustment from “Camera Adjustment 1” to “Camera Adjustment 8” starts.
- ④ **Reboot** button  
When this button is clicked, the camera is rebooted.
- ⑤ **Release Data Setting** button  
The data setting at the adjustment is cancelled.  
During the data setting, the button color changes from “white” to “red”. When the data setting is cancelled, the button color returns to “white”.  
(Use this button when an error occurred in the camera adjustment. If the adjustment completed successfully, the data setting is automatically cancelled and the button color returns to “white”.)

### 1-6-2. Adjustment Items of Camera System Adjustment

The adjustment items of camera system adjustment are as listed in Table 6-1-4. The Automatic Adjustment Program divides the adjustment items into eight, camera adjustment 1 to 8. Clicking either CAMERA Adjustment Start button allows the adjustment item which corresponds to that button to be executed.

The adjustment conditions of the subject and filter vary depending on which item is adjusted. The Adjustment Program displays an instruction for the subject and filter as a message during the adjustment.

| Button Name         | Adjustment                               | Subject                                                                                    | Adjusting Address |      |                                                                                        |
|---------------------|------------------------------------------|--------------------------------------------------------------------------------------------|-------------------|------|----------------------------------------------------------------------------------------|
|                     |                                          |                                                                                            | Block             | Page | Address                                                                                |
| CAMERA Adjustment 1 | Hall Adj.                                | Not required                                                                               | 11                | 61   | 0E00 to 0E04, 0E08, 0E09                                                               |
|                     | Auto Orientation Adj.                    |                                                                                            | 11                | 61   | 0E0A to 0E0D                                                                           |
| CAMERA Adjustment 2 | Flange Back Adj.                         | Siemens star chart with ND filter for minipattern box (Note) or Flange back adjustment jig | 11                | 61   | 069C to 069F, 06CC, 06D8 to 06DF, 0796 to 079D, 0F1C to 0F1D, 0F20, 0F24, 0F26 to 0F53 |
| CAMERA Adjustment 3 | Flange Back Check                        | Siemens star (1.0m from front the lens) (Illuminance: 200 to 400 lux)                      | -                 | -    | -                                                                                      |
| CAMERA Adjustment 4 | F No. Compensation                       | Clear chart (Standard picture frame)                                                       | 11                | 61   | 095A to 095F                                                                           |
|                     | Measure Gain, LV Adj.                    |                                                                                            | 11                | 61   | 0961 to 0968                                                                           |
|                     | Mechanical Shutter Adj.                  |                                                                                            | 11                | 61   | 0980 to 09AD                                                                           |
| CAMERA Adjustment 5 | AWB 3200K-5800K Standard Data Input      | 9 color chart (Standard picture frame)                                                     | 11                | 61   | 0C00 to 0C21, 0C24 to 0C49                                                             |
|                     | AWB 3200K-5800K Check                    |                                                                                            | -                 | -    | -                                                                                      |
|                     | Color Reproduction Adj. & Check          |                                                                                            | 11                | 61   | 0C50 to 0C57                                                                           |
|                     | CCD White Defect Compensation Check      | Clear chart (Standard picture frame)                                                       | 11                | 61   | 0200 to 03FF                                                                           |
|                     | CCD Black Defect Compensation Check      |                                                                                            | 11                | 61   | 0000 to 01FF                                                                           |
| CAMERA Adjustment 6 | Strobe Adj.                              | Flash adjustment box (50 cm)                                                               | 11                | 61   | 0C72 to 0C79                                                                           |
| CAMERA Adjustment 7 | Auto Focus Illumination Check            | Flash adjustment box (50 cm)                                                               | 11                | 61   | 0F10 to 0F15                                                                           |
| CAMERA Adjustment 8 | Angular Velocity Sensor Sensitivity Adj. | Not required                                                                               | 11                | 61   | 0E10, 0E11                                                                             |

**Note:** Dark Siemens star chart.

**Table 6-1-4**

### 1-6-3. Adjusting Method

#### 1. CAMERA Adjustment 1

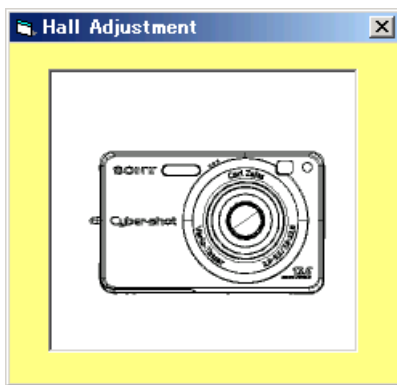
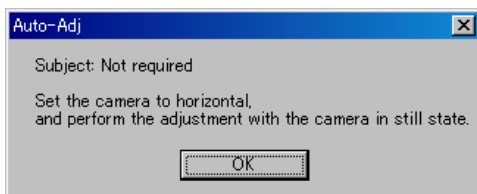
**Note:** There is no magnetic substance within around 8 cm of the camera.

##### [Automatic Adjustment Program execution items and sequence]

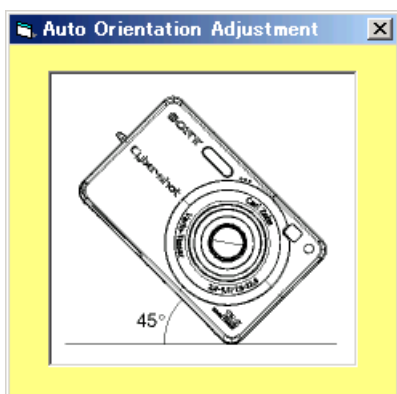
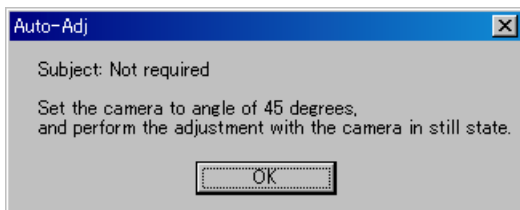
1. Data Setting during Camera Adj.
2. Hall Adj.
3. Auto Orientation Adj.
4. Release of Data Setting during Camera Adj.

##### [Adjusting method]

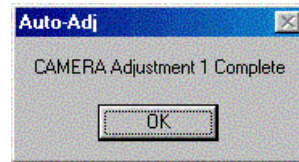
- 1) Click the **[Start]** button of CAMERA Adjustment 1.
- 2) The Automatic Adjustment Program executes the “1. Data Setting during Camera Adj.”.
- 3) Upon successful completion of “1. Data Setting during Camera Adj.”, the following message and screen are displayed. Set the camera in accordance with the message.



- 4) If the **[OK]** button is clicked, “2. Hall Adj.” will be executed.
- 5) Upon successful completion of “2. Hall Adj.”, the following message and screen are displayed. Set the camera in accordance with the message.



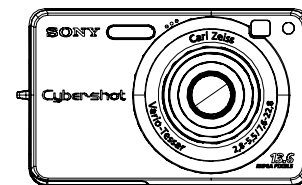
- 6) If the **[OK]** button is clicked, “3. Auto Orientation Adj.” and “4. Release of Data Setting during Camera Adj.” will be executed.
- 7) Upon successful completion of all items of the CAMERA Adjustment 1, the following message is displayed. Click the **[OK]** button.



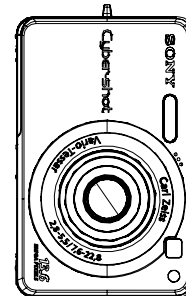
##### Processing after Completing Adjustment:

##### [Checking method]

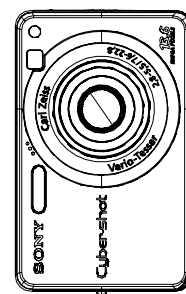
- 1) [HOME screen] → [Settings] → [Shooting Settings] → [Shooting Settings 2] → Select [Auto Orientation] to “ON”.
- 2) Shoot with the set in respective positions (a) to (c) shown below.
- 3) Check the pictures in the Playback mode to confirm that the pictures are rotated correctly.



(a)



(b)



(c)

Fig. 6-1-16

## 2. CAMERA Adjustment 2

### [Automatic Adjustment Program execution items and sequence]

1. Data Setting during Camera Adj.
2. Flange Back Adj.
3. Release of Data Setting during Camera Adj.

#### Preparation of Flange Back Adj.

##### (Using the minipattern box)

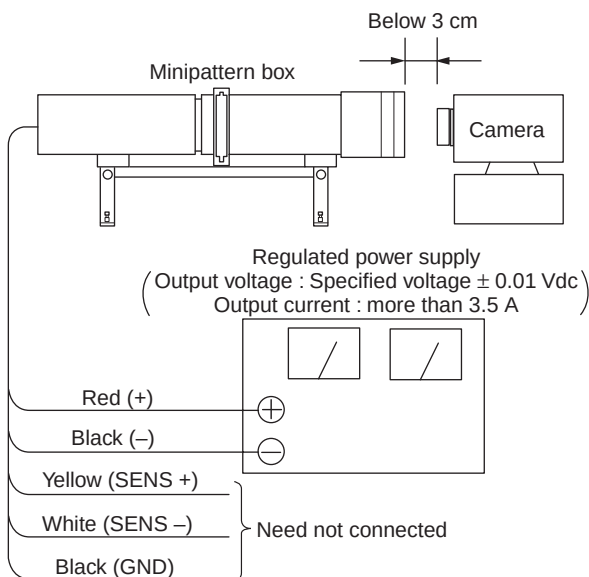
- 1) The minipattern box is installed as shown in the following figure.

**Note 1:** The attachment lenses are not used.

**Note 2:** Take care not to hit the mini-pattern box when extending the lens.

- 2) Install the minipattern box so that the distance between it and the front of lens of camera is less than 3 cm.
- 3) Make the height of minipattern box and the camera equal.
- 4) Check the output voltage of the regulated power supply is the specified voltage  $\pm 0.01$  Vdc.
- 5) Check that the center of Siemens star chart meets the center of shot image screen with the zoom lens at TELE end and WIDE end respectively.

**Specified voltage:** The specified voltage varies according to the minipattern box, so adjust the power supply output voltage to the specified voltage written on the sheet which is supplied with the minipattern box.



**Fig. 6-1-17**

#### Preparation of Flange Back Adj.

##### (Using the flange back adjustment jig)

##### (Illuminance: about 300 lux)

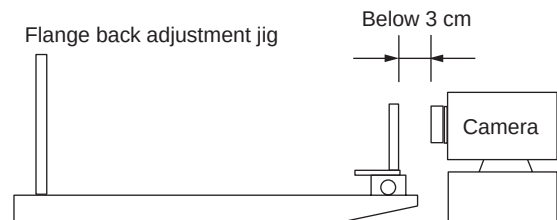
**Note 3:** When using the flange back adjustment jig, take care of the following points:

- For the illumination, use a light source such as an incandescent lamp or inverter type fluorescent light free from flickering.
- Do not make an adjustment in the environment where fluorescent lamp flickering occurs even if the illuminance can be ensured with the room illumination only. Use an incandescent lamp or inverter type fluorescent light at a place free from the influence of room illumination.

- 1) Install the flange back adjustment jig so that the distance between it and the front of lens of camera is less than 3 cm.

**Note 4:** Take care not to hit the flange back adjustment jig when extending the lens.

- 2) Make the height of flange back adjustment jig and the camera equal.
- 3) Check that the center of chart meets the center of shot image screen with the zoom lens at TELE end and WIDE end respectively.



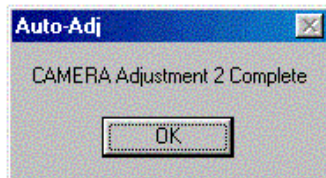
**Fig. 6-1-18**

#### [Adjusting method]

- 1) Click the **[Start]** button of the CAMERA Adjustment 2.
- 2) The Automatic Adjustment Program executes “1. Data Setting during Camera Adj.”.
- 3) Upon successful completion of the “1. Data Setting during Camera Adj.”, the following message is displayed. Set the subject by referring to “Preparation of Flange Back Adj.”.



- 4) If the **[OK]** button is clicked, “2. Flange Back Adj.” and “3. Release of Data Setting during Camera Adj.” will be executed.
- 5) Upon successful completion of all items of the CAMERA Adjustment 2, the following message is displayed. Click the **[OK]** button.



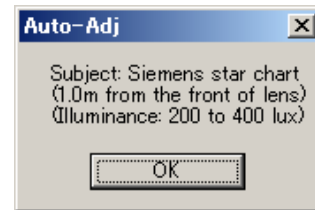
### 3. CAMERA Adjustment 3

#### [Automatic Adjustment Program execution items and sequence]

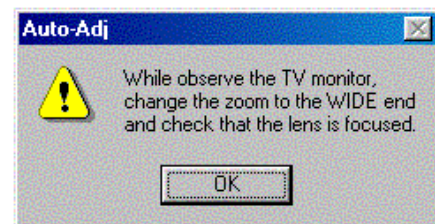
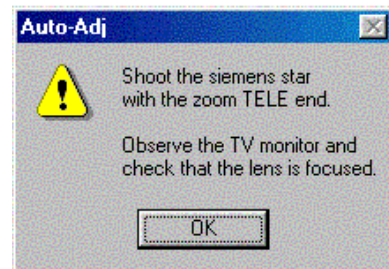
1. Data Setting during Camera Adj.
2. Flange Back Check
3. Release of Data Setting during Camera Adj.

#### [Adjusting method]

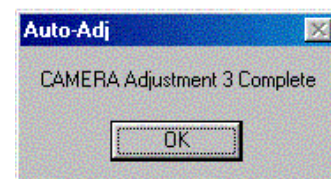
- 1) Click the **[Start]** button of the CAMERA Adjustment 3.
- 2) The Automatic Adjustment Program executes “1. Data Setting during Camera Adj.”.
- 3) Upon successful completion of the “1. Data Setting during Camera Adj.”, the following message is displayed. Set the subject in accordance with the message.



- 4) Click the **[OK]** button is clicked, “2. Flange Back Check” is executed. The following messages are displayed, and then operate the camera to make a check in accordance with the messages.



- 5) Upon completion of “2. Flange Back Check”, “3. Release of Data Setting during Camera Adj.” is executed.
- 6) Upon successful completion of all items of the CAMERA Adjustment 3, the following message is displayed. Click the **[OK]** button.



#### 4. Picture Frame Setting (Standard Picture Frame)

In the "CAMERA Adjustment 4 and CAMERA Adjustment 5", set the picture frame so as to attain the positions shown in the following figure when shooting the 9 colors chart.

##### Check on the oscilloscope

Measurement Point: Video terminal of USB, A/V, DC IN cable for multi-use terminal (75  $\Omega$  terminated)

##### 1. Horizontal period

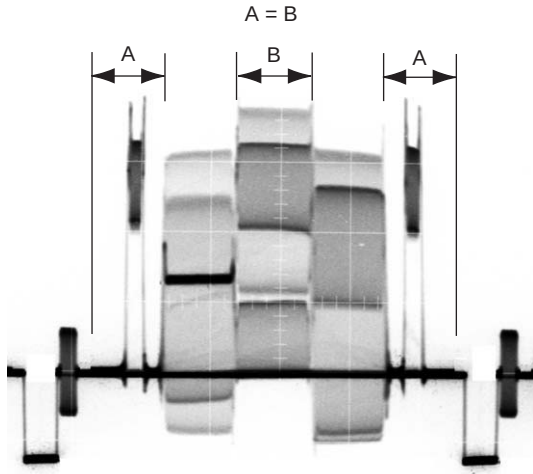


Fig. 6-1-19

##### 2. Vertical period

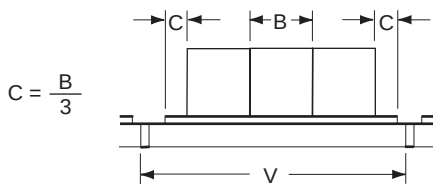


Fig. 6-1-20

##### Check on the monitor TV or the LCD screen

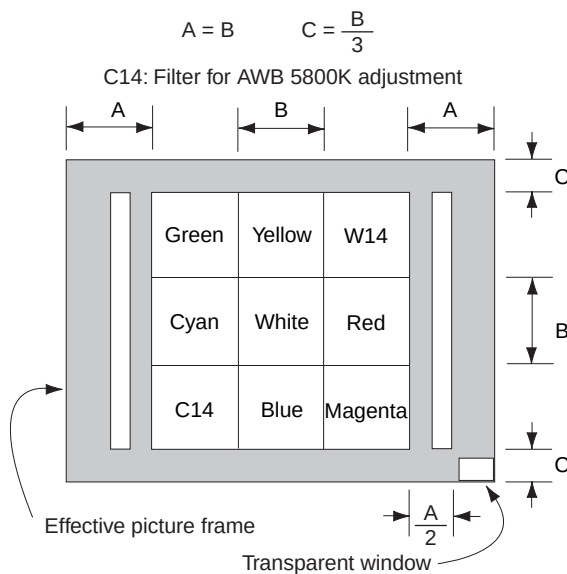


Fig. 6-1-21

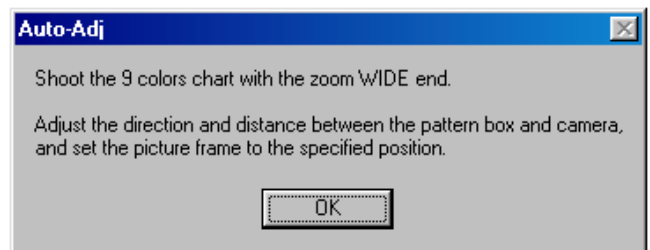
#### 5. CAMERA Adjustment 4

##### [Automatic Adjustment Program execution items and sequence]

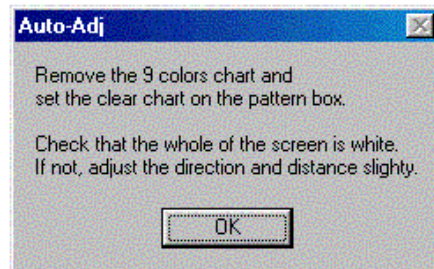
1. Data Setting during Camera Adj.
2. Picture Frame Setting
3. F No. Compensation
4. Measure Gain, LV Adj.
5. Mechanical Shutter Adj.
6. Release of Data Setting during Camera Adj.

##### [Adjusting method]

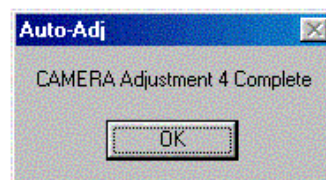
- 1) Click the **[Start]** button of the CAMERA Adjustment 4.
- 2) The Automatic Adjustment Program executes the "1. Data Setting during Camera Adj."
- 3) Upon successful completion of "1. Data Setting during Camera Adj.", "2. Picture Frame Setting" is executed. The following message is displayed, and then referring to Fig. 6-1-19 to Fig. 6-1-21, set the subject and click the **[OK]** button.



After that, the next message is displayed. Then, change the chart in accordance with the message.



- 4) Click the **[OK]** button, and the items from "3. F No. Compensation." to "6. Release of Data Setting during Camera Adj." will be executed.
- 5) Upon successful completion of all items of the CAMERA Adjustment 4, the following message is displayed. Click the **[OK]** button.



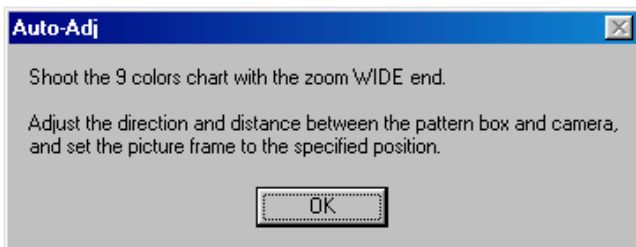
## 6. CAMERA Adjustment 5

### [Automatic Adjustment Program execution items and sequence]

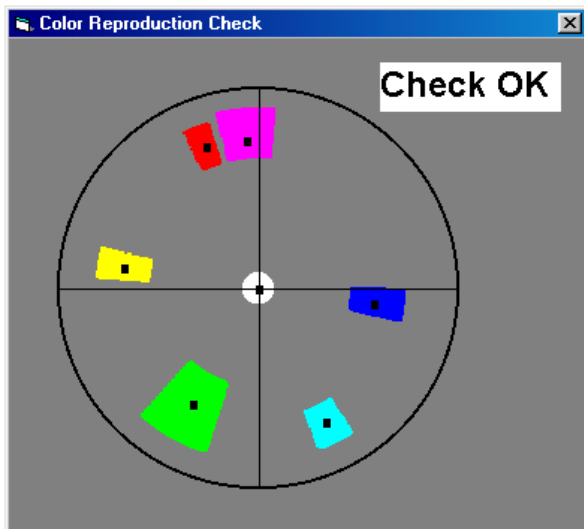
1. Data Setting during Camera Adj.
2. Picture Frame Setting
3. AWB 3200K-5800K Standard Data Input
4. AWB 3200K-5800K Check
5. Color Reproduction Adj. & Check
6. CCD White Defect Compensation Check
7. CCD Black Defect Compensation Check
8. Release of Data Setting during Camera Adj.

### [Adjusting method]

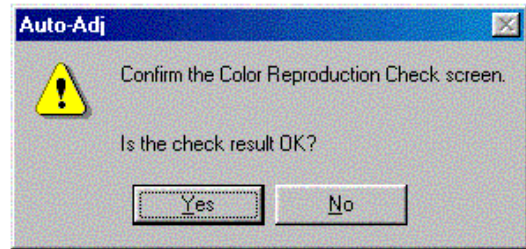
- 1) Click the **[Start]** button of the CAMERA Adjustment 5.
- 2) The Automatic Adjustment Program executes the “1. Data Setting during Camera Adj.”.
- 3) Upon successful completion of “1. Data Setting during Camera Adj.”, “2. Picture Frame Setting” is executed. The following message is displayed, and then referring to Fig. 6-1-17 to Fig. 6-1-19, set the subject and click the **[OK]** button.



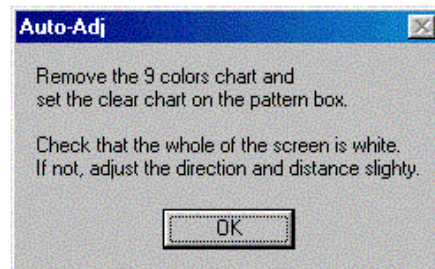
- 4) Click the **[OK]** button, and the “3. AWB 3200K-5800K Standard Data Input” and “4. AWB 3200K-5800K Check” will be executed.
- 5) After that, “5. Color Reproduction Adj. & Check” will be executed. Upon completion of adjustment, the check result is displayed on the Color Reproduction Check screen.



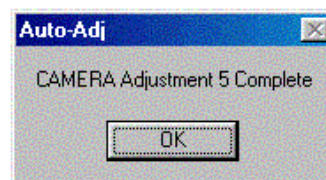
At this time, the following message is displayed, and click the **[Yes]** button if the check result display at the upper right of Color Reproduction Check screen is OK, or the **[No]** button if NG.



- 6) Upon successful completion of “5. Color Reproduction Adj. & Check”, the following message is displayed. Change the chart in accordance with the message.



- 7) Click the **[OK]** button, and the items from “6. CCD White Defect Compensation Check” to “8. Release of Data Setting during Camera Adj.” will be executed.
- 8) Upon successful completion of all items of the CAMERA Adjustment 5, the following message is displayed. Click the **[OK]** button.



## 7. CAMERA Adjustment 6

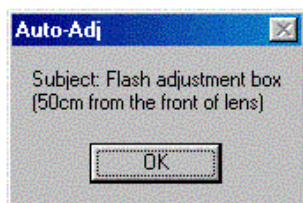
**Note:** “CAMERA Adjustment 6” is available only once after the power is turned on. If the adjustment is retried, turn off the power and turn on again.

### [Automatic Adjustment Program execution items and sequence]

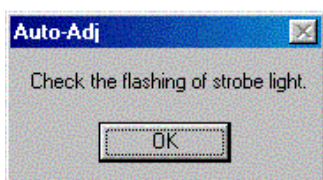
1. Data Setting during Camera Adj.
2. Strobe Adj.
3. Release of Data Setting during Camera Adj.

### [Adjusting method]

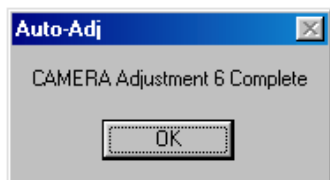
- 1) Click the **[Start]** button of CAMERA Adjustment 6.
- 2) The Automatic Adjustment Program executes the “1. Data Setting during Camera Adj.”.
- 3) Upon successful completion of the “1. Data Setting during Camera Adj.”, the following message is displayed. Set the subject in accordance with the message.  
(For the Flash adjustment box, refer to “5. Preparing the Flash Adjustment Box” (see page 6-7).)



- 4) Press the **[OK]** button, and the “2. Strobe Adj.” will be executed.
- 5) During execution of “2. Strobe Adj.”, the following message is displayed. After checking the flashing of strobe light, click the **[OK]** button. (This message is displayed 2 times during execution of adjustment.)



- 6) Upon successful completion of “2. Strobe Adj.”, “3.Release of Data Setting during CAMERA Adj.” is executed.
- 7) Upon successful completion of all items of the CAMERA Adjustment 6, the following message is displayed. Click the **[OK]** button.



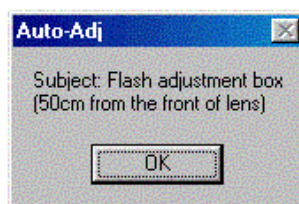
## 8. CAMERA Adjustment 7

### [Automatic Adjustment Program execution items and sequence]

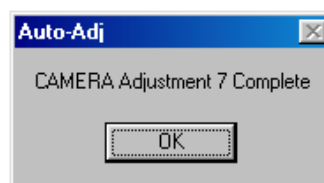
1. Data Setting during Camera Adj.
2. Auto Focus Illumination Check
3. Release of Data Setting during Camera Adj.

### [Adjusting method]

- 1) Click the **[Start]** button of CAMERA Adjustment 7.
- 2) The Automatic Adjustment Program executes the “1. Data Setting during Camera Adj.”.
- 3) Upon successful completion of the “1. Data Setting during Camera Adj.”, the following message is displayed. Set the subject in accordance with the message.  
(For the Flash adjustment box, refer to “5. Preparing the Flash Adjustment Box” (see page 6-7).)



- 4) Press the **[OK]** button, and the “2. Auto Focus Illumination Check” will be executed.
- 5) Upon successful completion of the “2. Auto Focus Illumination Check”, the “3. Release of Data Setting during Camera Adj.” will be executed successively.
- 6) Upon successful completion of all items of the CAMERA Adjustment 7, the following message is displayed. Click the **[OK]** button.



## 9. CAMERA Adjustment 8

Perform this adjustment only when replacing the angular velocity sensor or lens block. When the microprocessor, circuit etc. is damaged, don't perform this adjustment but check the operations only.

### 9-1. Precaution before adjustment

Before adjustment, read the following data and record them.  
When the SY-199 board is replaced, read out the data before the replacement, and record them.

#### Reading out method:

- 1) Read out the data in Block:11, Page:61, Address:0E10, and name this as Dp.
- 2) Read out the data in Block:11, Page:61, Address:0E11, and name this as Dy.

**Note:** The previous data (Dp, Dy) can be read from the SY-199 board and it can be saved in the PC as a file by using the Automatic Adjustment Program. (Refer to “9-4. Angular Velocity Sensor Sensitivity Adjustment”)

### 9-2. Preparation when the angular velocity sensor or the SY-199 board is replaced

Note down the sensitivity displayed on the angular velocity sensor of the repair parts. At this time, note down also to which board it was attached to.

Be sure to check because if attached incorrectly, the screen will vibrate up and down or left and right during the steady shot operations.

#### Precautions on the Parts Replacement

The PITCH sensor and the YAW sensor are different parts.

#### Precautions on Angular Velocity Sensor

The sensor incorporates a precision oscillator. Handle it with care as if it dropped, the balance of oscillator will be disrupted and operations will not be performed properly.

|                    |            |
|--------------------|------------|
| Adjustment Block   | 11         |
| Adjustment Page    | 61         |
| Adjustment Address | 0E10, 0E11 |

**Note:** The sensor sensitivity of SE501 and SE502 of the SY-199 board is written only repair parts.

#### Preparation:

- 1) Read the PITCH sensor (SY-199 board SE502) sensitivity written on repair parts, and named this as S<sub>502</sub>.
- 2) Read the YAW sensor (SY-199 board SE501) sensitivity written on repair parts, and named this as S<sub>501</sub>.

#### SY-199 BOARD (SIDE B)

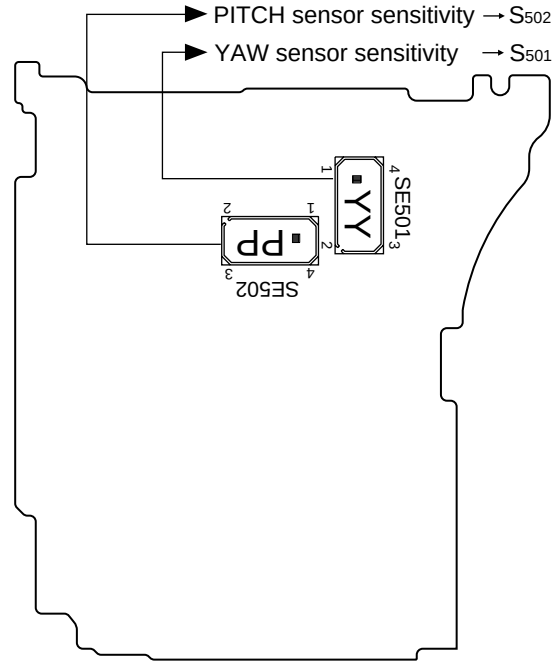


Fig. 6-1-22

#### How to read the sensitivity data of angular velocity sensor

With the pins 1 and 2 of angular velocity sensor placed in the lower position, read the data value.

#### Description example :

For the sensor sensitivity value 60

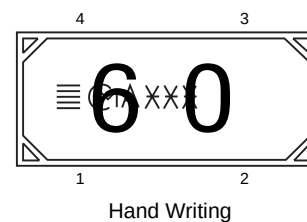


Fig. 6-1-23

### 9-3. Preparation when the lens is replaced

Note down the PITCH/YAW data on the replacement lens for repair.

|                    |            |
|--------------------|------------|
| Adjustment Block   | 11         |
| Adjustment Page    | 61         |
| Adjustment Address | 0E10, 0E11 |

**Note:** The PITCH/YAW data of lens is written only repair parts.

#### Preparation:

- 1) Read the PITCH data written on repair parts, and named this as L<sub>1</sub>.
- 2) Read the YAW data written on repair parts, and named this as L<sub>2</sub>.

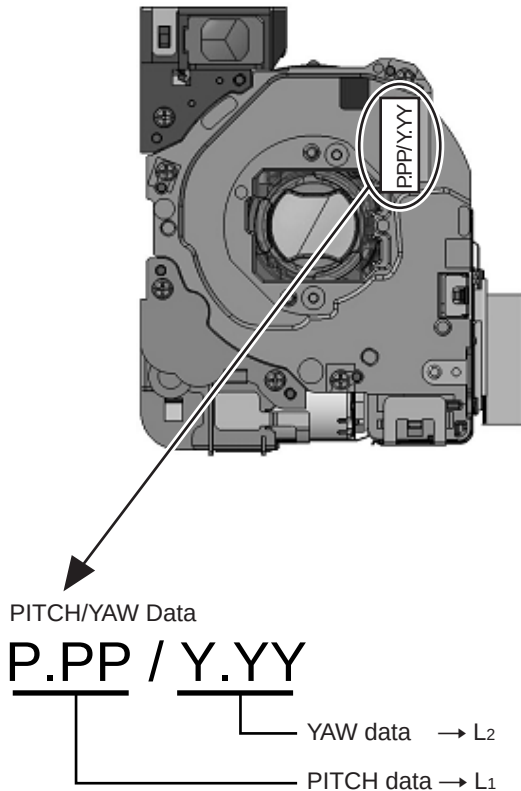
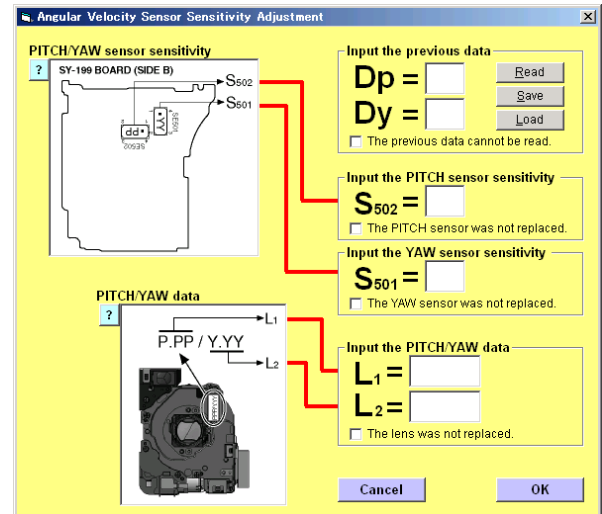


Fig. 6-1-24

### 9-4. Angular Velocity Sensor Sensitivity Adjustment [Adjusting method]

- 1) Click the **[Start]** button of Camera Adjustment 8, and the following screen will appear.



The buttons on this screen provide the following functions.

**[Read]** button: Reads and displays the previous data (Dp, Dy) from the SY-199 board.

**[Save]** button: Saves the previous data (Dp, Dy) in the PC as a file.

**[Load]** button: Loads and displays the previous data (Dp, Dy) from the file saved in the PC.

- Default file name is as follows:

DSC-W300\_DpDyDT\_XXXXXXXX\_YYYYMMDD.dat

\_\_\_\_\_ Date  
\_\_\_\_\_ USB serial number  
\_\_\_\_\_ Data name

- 2) Input the previous data (Dp, Dy) into screen.  
If the previous data cannot be read, give a check to the checkbox at lower left of screen.
- 3) Input the sensitivity of respective sensors (S<sub>501</sub>, S<sub>502</sub>) read at "Preparation" into the screen.  
If only either sensor was replaced, give a check to the checkbox for the sensor not replaced at the lower left of the screen.
- 4) Input the PITCH/YAW data of lens (L<sub>1</sub>, L<sub>2</sub>) read at "Preparation" into the screen.  
If the lens was not replaced, give a check to the checkbox at the lower left of the screen.

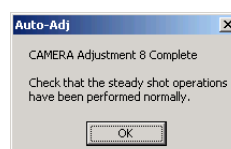
**Note:** In order to get the correct value for the adjustment, it is necessary to input a decimal point ("period" or "comma") as same format as windows setting because "Numeric Value Format" depend on the language.

(F:123 456 789,00, GB:123,456,789.00,

D:123.456.789,00)

"Numeric Value Format" setting is available in "Control Panel"- "Regional and Language Options".

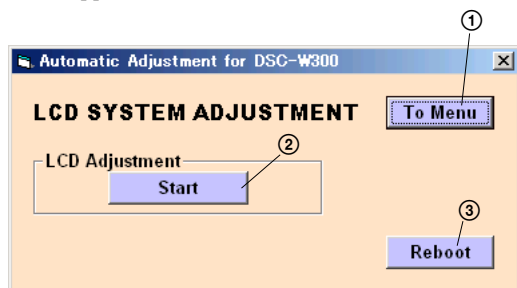
- 5) Click the **[OK]** button, and the adjustment data is then calculated from the sensor sensitivity value and the calculation result is written to the memory in the camera.
- 6) Upon successful completion of the data writing, the following screen will appear. Check that the steady shot function operates normally.



## 1-7. LCD SYSTEM ADJUSTMENTS

### 1-7-1. Function of Each Button on LCD System Adjustment Screen

Click the **LCD SYSTEM ADJUSTMENT** button on the Main Menu screen, and the “LCD SYSTEM ADJUSTMENT” screen in Fig. 6-1-25 will appear.



**Fig. 6-1-25**

- ① **To Menu** button  
Return to the main menu.
- ② **Start** button  
“LCD Adjustment” starts.
- ③ **Reboot** button  
When this button is clicked, the camera is rebooted.

### 1-7-2. Adjustment Items of LCD System Adjustment

The adjustment items of LCD system adjustment are as listed in Table 6-1-5. The Automatic Adjustment Program executes the adjustment items if the LCD Adjustment Start button is clicked.

| Button Name | Adjustment         | Adjusting Address |      |            |
|-------------|--------------------|-------------------|------|------------|
|             |                    | Block             | Page | Address    |
| LCD         | V-COM Adj.         | 11                | 60   | 0401       |
| Adjustment  | White Balance Adj. | 11                | 60   | 0362, 0363 |

**Table 6-1-5**

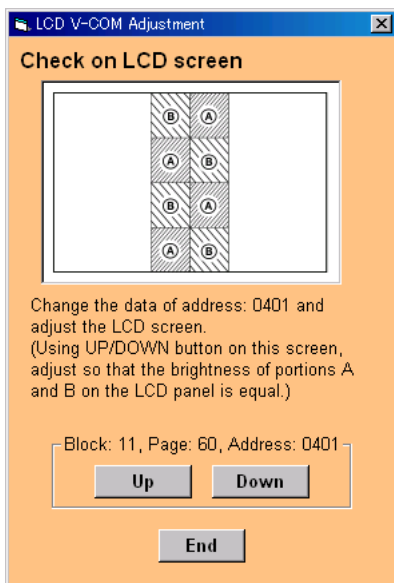
### 1-7-3. Adjusting Method

#### [Automatic Adjustment Program execution items and sequence]

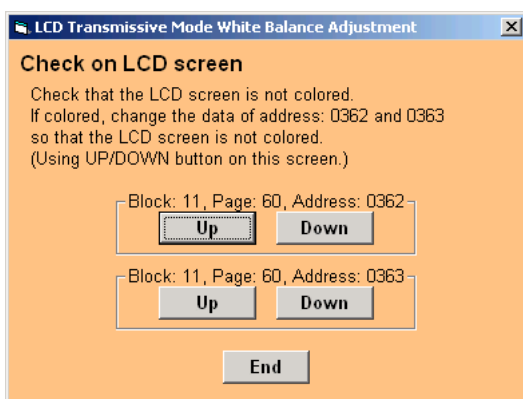
1. Data Setting during LCD Adj.
2. V-COM Adj.
3. White Balance Adj.

#### [Adjusting method]

- 1) Click the **[Start]** button of the LCD Adjustment.
- 2) The Automatic Adjustment Program executes “1. Data Setting during LCD Adj.”.
- 3) Upon successful completion of the “1. Data Setting during LCD Adj.”, the following screen is displayed during the execution of “2. V-COM Adj.”. Using the **[Up]/[Down]** button on the screen, adjust so that the brightness of portions A and B on the LCD panel is equal. After the adjustment, click the **[End]** button in the screen.



- 4) If the **[End]** button is clicked, the following screen is displayed during the execution of “3. White Balance Adj.”. Check that the LCD screen is not colored. If colored, using the **[Up]/[Down]** button on the screen, adjust so that the LCD screen is not colored. After the adjustment, click the **[End]** button in the screen.



- 5) Upon successful completion of all item the LCD Adjustment, the following message is displayed. Click the **[OK]** button.



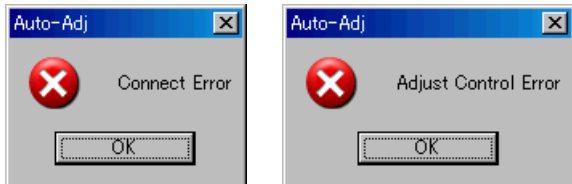
## 1-8. ERROR

In the case of an error during the execution of adjustment, the Automatic Adjustment Program interrupts the processing at that point, and displays an error message, and then terminates the program execution there.

### 1-8-1. Error Message

When an error message is displayed, perform the remedy given below, and then retry adjustment. If the error message is displayed though the remedy was performed, the circuits will be faulty.

#### 1. Connect Error, Adjust Control Error



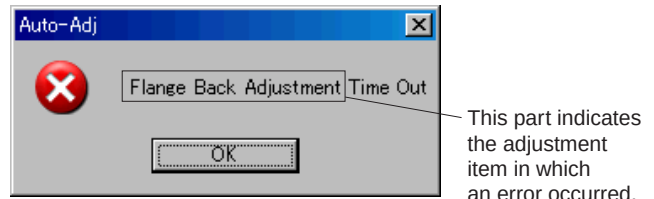
|         |                                                                                                                                                                                                                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symptom | USB communication with the set is abnormal.                                                                                                                                                                                                                                                                      |
| Cause   | <ul style="list-style-type: none"> <li>• USB cable is not inserted tightly.</li> <li>• Power supply is not installed correctly.</li> <li>• Communication with SeusEX is abnormal.</li> </ul>                                                                                                                     |
| Remedy  | <ul style="list-style-type: none"> <li>• Disconnect the USB cable once, and then re-connect it tightly and check that the set is in "USB Mode".</li> <li>• Install the power supply correctly.</li> <li>• Start the SeusEX and click the [Connect] to check that the connection state is established.</li> </ul> |

#### 2. RESET the CAMERA and Try Again



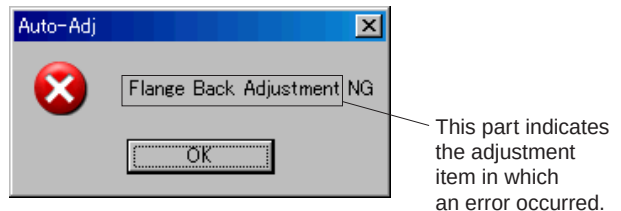
|         |                                         |
|---------|-----------------------------------------|
| Symptom | The camera is not ready for adjustment. |
| Cause   | Data error exists in the camera.        |
| Remedy  | Reset the camera.                       |

#### 3. Adjustment Time Out



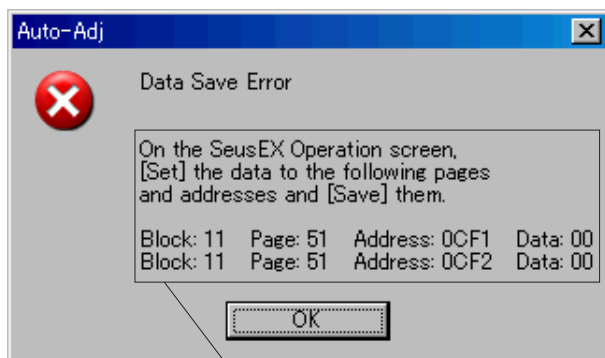
|         |                                                                                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Symptom | Adjustment does not finish within the specified time.                                                                                     |
| Cause   | <ul style="list-style-type: none"> <li>• Adjustment conditions are wrong.</li> <li>• Data error exists in the camera.</li> </ul>          |
| Remedy  | <ul style="list-style-type: none"> <li>• Check that the conditions such as a subject are correct.</li> <li>• Reset the camera.</li> </ul> |

#### 4. Adjustment NG



|         |                                                                                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Symptom | The adjusted data does not become the specified value.                                                                                    |
| Cause   | <ul style="list-style-type: none"> <li>• Adjustment conditions are wrong.</li> <li>• Data error exists in the camera.</li> </ul>          |
| Remedy  | <ul style="list-style-type: none"> <li>• Check that the conditions such as a subject are correct.</li> <li>• Reset the camera.</li> </ul> |

## 5. Data Save Error



How to cancel the data setting during adjustment is display here.

|         |                                                                                                                                                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symptom | data cannot be saved normally. (The data setting during adjustment cannot be cancelled)                                                                                                                                                                                                                        |
| Cause   | <ul style="list-style-type: none"> <li>• Data writing to the flash memory failed.</li> <li>• Connection is faulty.</li> <li>• Power supply is not installed correctly.</li> </ul>                                                                                                                              |
| Remedy  | <ul style="list-style-type: none"> <li>• On the SeusEX Operation screen, [Set] the data to the pages and addresses displayed in the message, and [Save] them. (Cancel manually the data setting during adjustment.)</li> <li>• Check the connection.</li> <li>• Install the power supply correctly.</li> </ul> |

## 1-8-2. Precautions When an Error Occurred

The Automatic Adjustment Program sets the data for adjustment before the adjustment starts. Accordingly, if the adjustment terminates by an error, the data during the adjustment may be left in the camera.

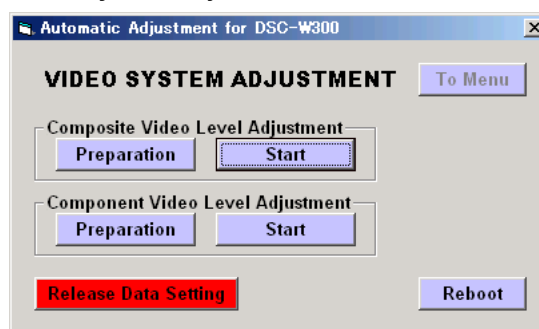
**Note 1:** With this data left in the camera, the camera will not operate normally.

In this case, the [Release Data Setting] button is displayed in “red” on the screen as shown figures below. Click the [Release Data Setting] button to cancel the data setting. When the data setting is cancelled, the button color becomes “white”.

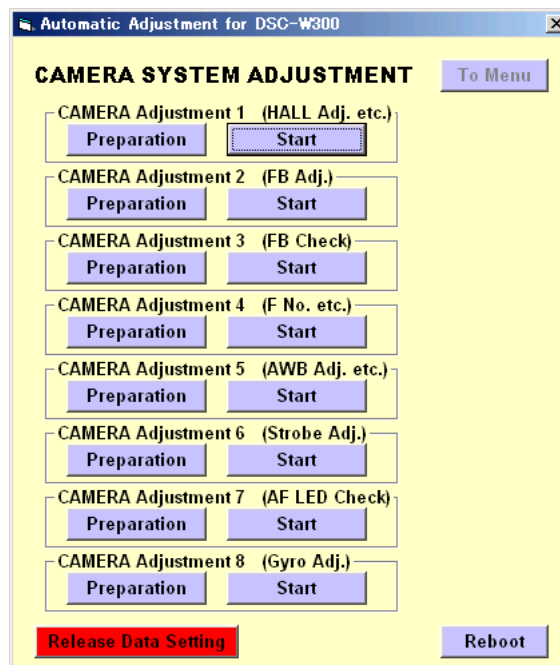
**Note 2:** When “Data Save Error” occurred, the [Release Data Setting] button is displayed in “white”.

To cancel the data setting, perform it on the SeusEX Operation screen. How to cancel the data setting is displayed in the error message.

## Video System Adjustment screen



## Camera System Adjustment screen



## 6-2. SERVICE MODE

### 2-1. APPLICATION FOR ADJUSTMENT (SeusEX)

The adjustment software (SeusEX) can change operational coefficients of signal processing, EVR data, etc. same as the adjustment remote commander. The SeusEX performs two-way communication between PC and camcorder using the USB terminal. The two-way communication result data can be written in the nonvolatile memory.

#### 1. Connection

- 1) Connect the HASP key to the USB terminal of the PC.
- 2) Connect the PC and camcorder with the USB cable.
- 3) Start the SeusEX on the PC.
- 4) Click **Connect** on the SeusEX screen. If the connection is normal, the SeusEX screen will be as shown in Fig. 6-2-1, indicating the “connected” state.

**Note:** The SeusEX will go in “disconnect” state, if the camcorder is turned off (for instance, by resetting the set). In such a case, click **Connect** on the SeusEX screen to restore the “connected” state.

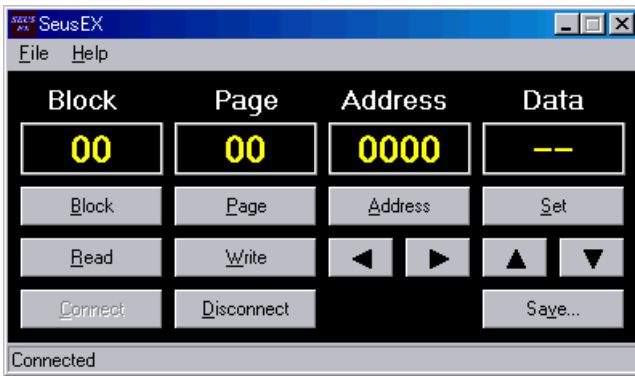


Fig. 6-2-1

#### 2. Operation

- Block change  
To change the block, click **Block** on the SeusEX screen and enter the block to be changed. The block is displayed in hexadecimal notation.
- Page change  
To change the page, click **Page** on the SeusEX screen and enter the page to be changed. The page is displayed in hexadecimal notation.
- Address change  
To change the address, click **Address** on the SeusEX screen and enter the address to be changed. The address is displayed in hexadecimal notation.
- Data change  
To change the data, click **Set** on the SeusEX screen and enter the data. The data is displayed in hexadecimal notation. This operation does not write the data to the nonvolatile memory.
- Data writing  
To write the data to the EEPROM, click **Write** on the SeusEX screen and enter the data value to be written. To write the data to the flash memory, change the data value using the **Set** on the SeusEX screen and then click **Save** to save the data.
- Data reading  
The data displayed on the SeusEX screen are the data values at the time when the pages and addresses were set, and they are not updated automatically. To check the data change, click **Read** on the SeusEX screen and update the displayed data.

## 2-2. SERVICE MODE

### 1. Function of Each Button on Service Mode Screen

Click the **SERVICE MODE** button on the Main Menu screen, and the “SERVICE MODE” screen in Fig. 6-2-2 will appear.

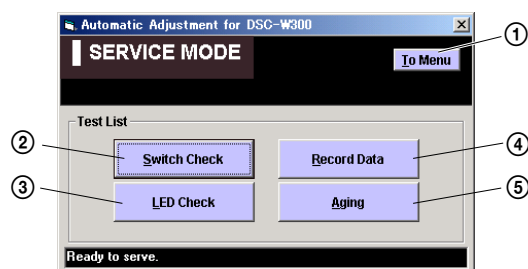
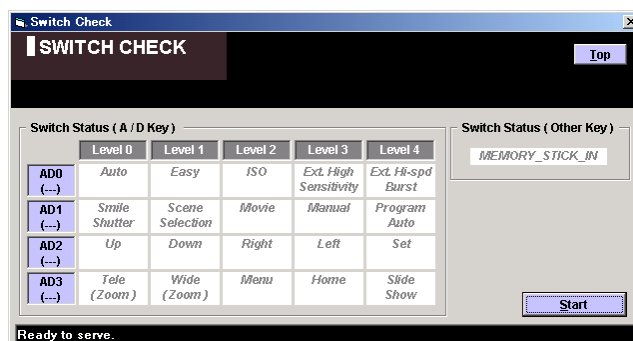


Fig. 6-2-2

- ① **To Menu** button  
Return to the main menu.
- ② **Switch Check** button  
“SWITCH CHECK” screen appears.
- ③ **LED Check** button  
“LED CHECK” screen appears.
- ④ **Record Data** button  
“RECORD DATA” screen appears.
- ⑤ **Aging** button  
“AGING” screen appears.

### 2. Switch Check

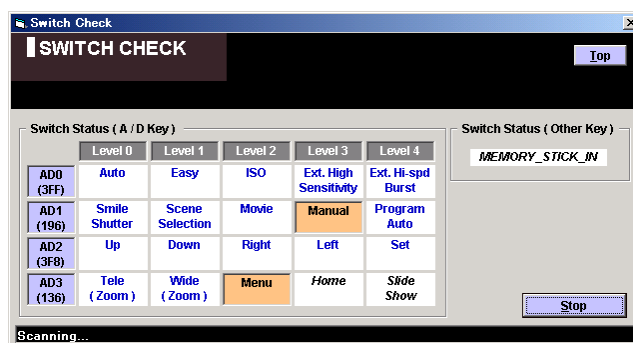
Click the **Switch Check** button on the SERVICE MODE screen, and the “SWITCH CHECK” screen will appear.



#### Using method:

Click the **Start** button, and the switch check will start.

During execution of switch check, the pressed switch is displayed in orange. Also, once the switch was pressed, its name characters change to blue in color.



### 3. LED Check

Click the **LED Check** button on the SERVICE MODE screen, and the “LED CHECK” screen will appear.

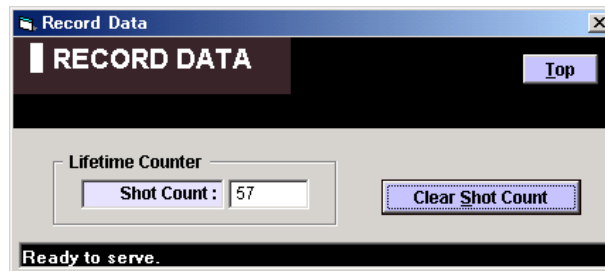


#### Using method:

LED ON or OFF can be controlled with **LED ON** or **LED OFF** button on the screen.

### 4. Record Data

Click the **Record Data** button on the SERVICE MODE screen, and the “RECORD DATA” screen will appear.



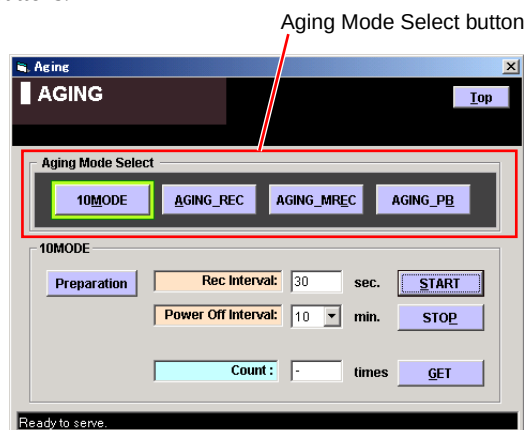
The record data such as Lifetime counter is displayed.

#### Initializing method:

Click the **Clear Shot Count** button on the screen, and the Lifetime Count data will be initialized.

## 5. Aging

On the AGING screen, various types of aging can be executed. Select the type of aging to be executed with the Aging Mode Select buttons.



Supply the power not from an AC adapter but from the fully charged battery and execute the aging until the battery end, so that its battery use time and how many pictures the camera can record/play back can be checked.

During the execution of aging, the camera automatically repeats recording/playback.

During the recording, if the recording memory capacity becomes full, its memory is automatically formatted.

Accordingly, when executing the aging, insert the Memory Stick into the camera.

### [Preparation]

For the aging, set the camera as follows:

#### Switch Setting

- 1) Mode Dial ..... Auto Adjustment

#### Home Items

- 1) AF Mode  
(Settings – Shooting Settings – Shooting Settings 1) .... Single

#### Menu Items

- 1) REC Mode ..... Normal

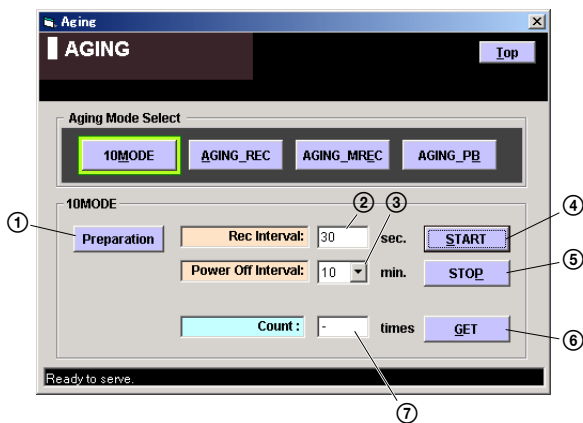
## (1) 10MODE

In the 10MODE, perform continuous recording in accordance with the CIPA Standard. When the 10MODE starts, the following operation is repeated until the battery end.

- 1) The zoom moves to the TELE end.
- 2) The camera records a still picture with a strobe light flashed at the maximum level.
- 3) The camera stands by for the time (sec.) set to the “Rec Interval”.
- 4) The zoom moves to the WIDE end.
- 5) The camera records a still picture without flashing a strobe light.
- 6) The camera stands by for the time (sec.) set to the “Rec Interval”.
- 7) Each time 10 still pictures are recorded, the power is turned off, and after the time (min.) set to the “Power Off Interval”, the power is turned on again.

At the start of 10MODE, the counter in the camera is initialized. This counter counts the number of recording times during the execution of 10MODE, and it holds the counts even during the power off.

## 10MODE screen and functions of each item



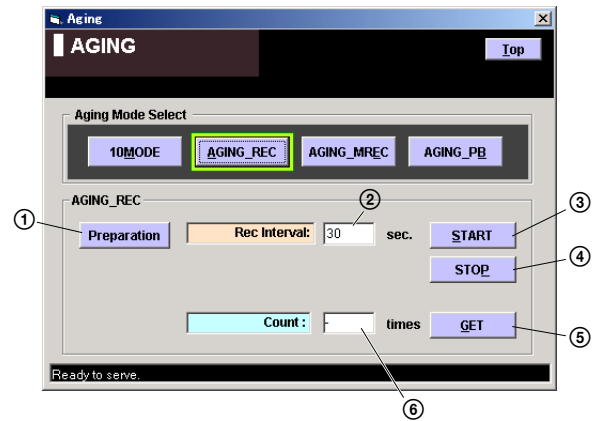
- ① **Preparation** button  
Notes for aging is displayed.
- ② **Rec Interval** setting and display  
The setting of recording interval (sec.) is displayed. The setting can be changed by entering a numeric value in this field.
- ③ **Power Off Interval** setting and display  
The setting of power off interval (min.) is displayed. The setting can be changed by entering a numeric value in this field.
- ④ **START** button  
The settings of parameters specified by the Rec Interval and Power Off Interval for execution of 10MODE are sent to the camera.  
The 10MODE starts when the camera power is turned off once and then on again.  
The settings for execution of 10MODE are held until the camera detects the battery end. They cannot be reset by the power off or RESET button. To interrupt the 10MODE, use the **STOP** button.
- ⑤ **STOP** button  
The settings for execution of 10MODE are reset.
- ⑥ **GET** button  
After the 10MODE was executed, the number of recording times is got from the counter in the camera. The result is displayed in the "Count" display field.
- ⑦ **Count** display  
The number of recording times got by the **GET** button is displayed.

## (2) AGING\_REC

In the AGING\_REC mode, still pictures are continuously recorded at the interval (sec.) set to the "Rec Interval".

At the start of AGING\_REC, the counter in the camera is initialized. This counter counts the number of recording times during the execution of AGING\_REC, and it holds the counts even during the power off.

## AGING\_REC screen and functions of each item



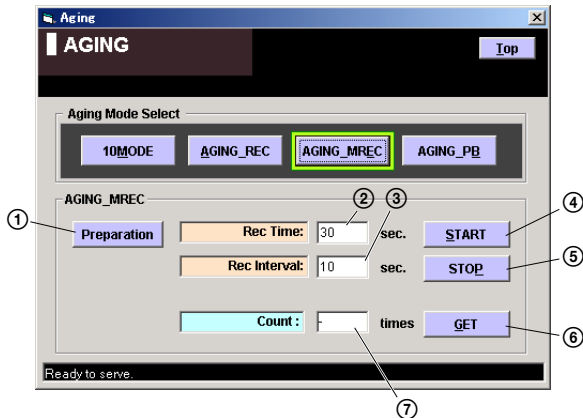
- ① **Preparation** button  
Notes for aging is displayed.
- ② **Rec Interval** setting and display  
The setting of recording interval (sec.) is displayed. The setting can be changed by entering a numeric value in this field.
- ③ **START** button  
The AGING\_REC starts. In the AGING\_REC mode, the aging starts immediately when the **START** button is clicked.  
At the battery end, the AGING\_REC stops. To interrupt the AGING\_REC, use the **STOP** button, though the AGING\_REC stops also by the power off or RESET button.
- ④ **STOP** button  
The AGING\_REC stops.
- ⑤ **GET** button  
After the AGING\_REC was executed, the number of recording times is got from the counter in the camera. The result is displayed in the "Count" display field.
- ⑥ **Count** display  
The number of recording times got by the **GET** button is displayed.

### (3) AGING\_MREC

In the AGING\_MREC mode, motion pictures are continuously recorded for the time (sec.) set to the “Rec Time” at the interval (sec.) set to the “Rec Interval”.

At the start of AGING\_MREC, the counter in the camera is initialized. This counter counts the number of recording times during the execution of AGING\_MREC, and it holds the counts even during the power off.

#### AGING\_MREC screen and functions of each item



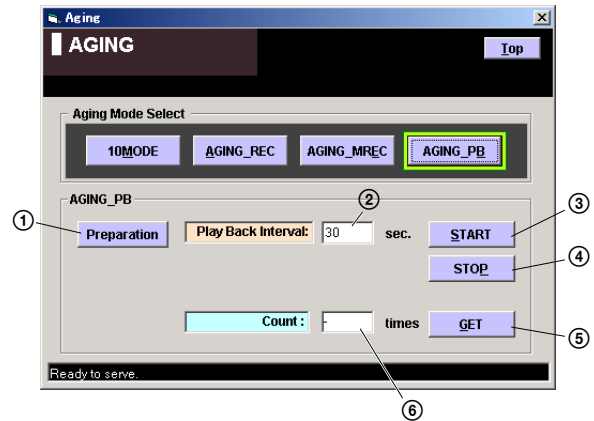
- ① **[Preparation]** button  
Notes for aging is displayed.
- ② **Rec Time** setting and display  
The setting of recording time (sec.) is displayed. The setting can be changed by entering a numeric value in this field.
- ③ **Rec Interval** setting and display  
The setting of recording interval (sec.) is displayed. The setting can be changed by entering a numeric value in this field.
- ④ **[START]** button  
The AGING\_MREC starts. In the AGING\_MREC mode, the aging starts immediately when the **[START]** button is clicked. At the battery end, the AGING\_MREC stops. To interrupt the AGING\_MREC, use the **[STOP]** button, though the AGING\_MREC stops also by the power off or RESET button.
- ⑤ **[STOP]** button  
The AGING\_MREC stops.
- ⑥ **[GET]** button  
After the AGING\_MREC was executed, the number of recording times is got from the counter in the camera. The result is displayed in the “Count” display field.
- ⑦ **Count** display  
The number of recording times got by the **[GET]** button is displayed.

### (4) AGING\_PB

In the AGING\_PB mode, pictures are continuously played back at the interval (sec.) set to the “Play Back Interval”.

At the start of AGING\_PB, the counter in the camera is initialized. This counter counts the number of play back times during the execution of AGING\_PB, and it holds the counts even during the power off.

#### AGING\_PB screen and functions of each item



- ① **[Preparation]** button  
Notes for aging is displayed.
- ② **Play Back Interval** setting and display  
The setting of play back interval (sec.) is displayed. The setting can be changed by entering a numeric value in this field.
- ③ **[START]** button  
The AGING\_PB starts. In the AGING\_PB mode, the aging starts immediately when the **[START]** button is clicked. At the battery end, the AGING\_PB stops. To interrupt the AGING\_PB, use the **[STOP]** button, though the AGING\_PB stops also by the power off or RESET button.
- ④ **[STOP]** button  
The AGING\_PB stops.
- ⑤ **[GET]** button  
After the AGING\_PB was executed, the number of play back times is got from the counter in the camera. The result is displayed in the “Count” display field.
- ⑥ **Count** display  
The number of play back times got by the **[GET]** button is displayed.

### 2-3. DATA BACKUP

With the “DATA BACKUP”, the adjustment data in the camera can be backed up in the PC as a file.

The adjustment data that can be backed up are as follows.

#### System Control Section:

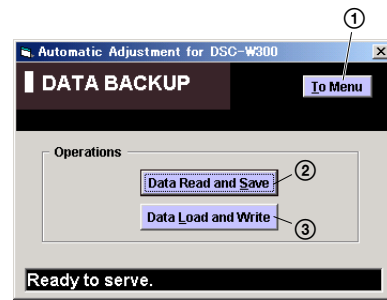
- 1) Video System Adjustments
- 2) LCD System Adjustments

#### Camera Control Section:

- 1) Camera System Adjustments

### 1. Function of Each Button on Data Backup Screen

Click the [DATA BACKUP] button on the Main Menu screen, and the “DATA BACKUP” screen in Fig. 6-2-3 will appear.



**Fig. 6-2-3**

- ① [To Menu] button  
Return to the main menu.
- ② [Data Read and Save] button  
Read the adjustment data from the camcorder and save them in PC as a file.  
Default file name is as follows:  
DSC-W300\_ADJBAK\_XXXXXXX\_YYYYMMDD.dat  

|  |  |  |                   |
|--|--|--|-------------------|
|  |  |  | Date              |
|  |  |  | USB serial number |
|  |  |  | Data name         |
- ③ [Data Load and Write] button  
Load the adjustment data from the file saved in PC and write them to the camcorder.

## Revision History

| Ver. | Date    | History                    | Contents                                                                                                                                                                                                                                                                                                                                                                                            | S.M. Rev. issued |
|------|---------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.0  | 2008.03 | Official Release           | —                                                                                                                                                                                                                                                                                                                                                                                                   | —                |
| 1.1  | 2008.05 | Revised-1<br>(A1 DI08-068) | Replace the previously issued SERVICE MANUAL 9-852-287-51 with this Manual. <ul style="list-style-type: none"> <li>• Change of Automatic Adjustment Program<br/>Version of Automatic Adjustment Program has been changed from Ver_1.0r01 into Ver_1.1r02.</li> <li>• Addition of Thai Model<br/>S.M. revised: <a href="#">Page 6-8</a>, <a href="#">Page 6-10</a></li> </ul>                        | Yes              |
| 1.2  | 2008.06 | Revised-2<br>(A2 DI08-113) | Replace the previously issued SERVICE MANUAL 9-852-287-52 with this Manual. <ul style="list-style-type: none"> <li>• Change of Automatic Adjustment Program<br/>Version of Automatic Adjustment Program has been changed from Ver_1.1r02 into Ver_1.2r03.</li> <li>• Correction of Destination Data Write<br/>S.M. revised: <a href="#">Page 6-8</a></li> </ul>                                     | Yes              |
| 1.3  | 2008.08 | Revised-3<br>(A3 DI08-225) | Replace the previously issued SERVICE MANUAL 9-852-287-53 with this Manual. <ul style="list-style-type: none"> <li>• Change of Automatic Adjustment Program<br/>Version of Automatic Adjustment Program has been changed from Ver_1.2r03 into Ver_1.3r04.</li> <li>• Correction of the “Setting of Adjustment Mode”<br/>S.M. revised: <a href="#">Page 6-8</a>, <a href="#">Page 6-9</a></li> </ul> | Yes              |
|      |         |                            |                                                                                                                                                                                                                                                                                                                                                                                                     |                  |